Refrigeration, Food Security, and Nutrition: Evidence from Ghana

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Abstract

Food insecurity in low-income countries is typically analysed through agricultural production and purchasing power, with little attention paid to downstream food distribution. We study the welfare consequences of disruptions in final-stage food storage. Our identification exploits appliance breakdowns in Ghana, comparing households with functioning and broken refrigerators purchased at the same time and at similar prices. Breakdowns increase perceived food insecurity by one-third and reduce consumption of animal-sourced foods, lowering intake of vitamin B12. These findings highlight household refrigeration as an underappreciated lever to improve diets and reduce widespread micronutrient deficiencies as electrification rapidly expands.

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I Introduction

The risk that food never reaches households is largely absent from standard analyses of food insecurity, which generally emphasize agricultural production and purchasing power while paying relatively little attention to frictions in supply chains and final-stage food distribution.

In this paper, we study the welfare consequences of disruptions to household food storage. We find that storage constraints at the point of final consumption substantially increase perceived food insecurity, reduce dietary diversity, and limit access to essential micronutrients. Our findings identify household food storage as an important, yet often overlooked, determinant of food access. This mechanism has global policy relevance. Nearly 900 million people face hunger and malnutrition worldwide (FAO et al., 2024), and while cold chains remain incomplete in many low-income countries, global electrification and appliance diffusion are expected to bring household refrigeration to roughly 800 million new households over the coming decade (IEA et al., 2023), creating an unprecedented opportunity to reduce food insecurity.

We use data from Ghana, where almost half of the population experiences moderate to severe food insecurity (GSS, 2022). Ghana is also an early adopter of grid connection in sub-Saharan Africa, where refrigerator ownership rose from 9.1 percent in 1993 to 41.9 percent in 2022 (ICF, 1993, 2022). Our identification strategy exploits plausibly exogenous variation in appliance breakdowns, conditional on appliance price and purchase year. Lacking access to a functioning refrigerator increases the probability of food insecurity by 33 percent. A key mechanism is reduced dietary diversity: households without refrigeration acquire 21 percent less meat, 18 percent less dairy, 39 percent fewer eggs, and 10 percent less fish. Total energy and macronutrient availability change little, indicating substitution rather than reductions in food intake. In contrast, refrigeration materially increases intakes of key micronutrients concentrated in animal-sourced foods, especially vitamin B12, which is essential for red blood cell formation, neurological function, and fetal development.

Estimation relies on an augmented inverse probability weighting (AIPW) estimator with LASSO-based covariate selection (Chernozhukov et al., 2018), allowing comparisons across similar households in the same location who acquired refrigerators at the same

time and at comparable prices. This approach is analogous to a matching estimator that adjusts for the main determinants of treatment assignment (in this case, whether the appliance broke down). The main identification risk is hidden bias. We address this concern with Rosenbaum's sensitivity analysis (Rosenbaum, 2002a,b, 2005), which tests how much unobserved heterogeneity in the odds of treatment would be required to overturn the results. Our estimates remain statistically significant even under levels of hidden bias that would need to be larger than the selection effects associated with appliance price or time since purchase. Placebo tests using unrelated broken appliances show no effects. Results also hold when focusing on households that received refrigerators as a gift, showing omitted variable biases linked to appliance selection are not driving our findings.

This paper contributes to three strands of literature. First, it contributes to the economics of food security by identifying household-level storage as a critical and commonly overlooked determinant of food insecurity and nutrition. Few studies have paid attention to the role of cold value chains and intermediaries in shaping household food access in LMICs (Barrett et al., 2022; Macchiavello et al., 2022). In contexts with high transport costs and poor storage, intermediaries have been shown to extract significant rents (Bergquist et al., 2020), reducing affordability for consumers. Suri and Udry (2022) also point to substantial heterogeneity in the returns to agricultural technologies in sub-Saharan Africa, which post-harvest bottlenecks may partly drive. Improving cold chains could help address these inefficiencies both upstream in the value chain and at the point of final consumption. Refrigeration could play a complementary role alongside existing interventions such as food fortification, supplementation, nutrition education, and direct food provision (Allen and Gillespie, 2001; De Marchis et al., 2019), as well as demand-side efforts like cash transfers aimed at improving food affordability (Bhalla et al., 2018; Burchi and Strupat, 2016; Debela et al., 2021; Tiwari et al., 2016). By improving access to micronutrient-rich foods, cold chains may help close persistent nutrition gaps that traditional interventions have struggled to fully address.[1]

Second, this paper contributes to the literature on infrastructure and energy access by distinguishing access to electricity from access to energy services and then analyzing its

[1]This aligns with the Global Cooling Pledge, endorsed by over 70 countries at the 2023 UN Climate Change Conference, which calls for National Cooling Adaptation Plans that incorporate access to cooling and a strengthening of cold chains to foster sustainable development.

direct welfare consequences. At the national level, electrification has been linked to gains in employment and productivity in countries like South Africa, India, Ghana, and Brazil (Dinkelman, 2011; Lipscomb et al., 2013; Ntsiful et al., 2024; Rud, 2012). However, Lee et al. (2020b) show that these gains are far from automatic and hinge on actual energy use. Barriers include low energy affordability (Dube and Ikhupuleng, 2003; Lee and Yuan, 2024; Nchofoung, 2024), supply-side constraints (Burgess et al., 2020), and, as this paper underscores, appliance usage.[2]

Targeting energy poverty more effectively may require shifting focus from grid access to the specific services that households need to improve welfare. By enabling safe storage and greater intake of micronutrient-rich foods, refrigerators help households adapt to dietary risk, just as air conditioners and heaters help them adapt to climate risk (Deschênes and Greenstone, 2011; Barreca et al., 2016; Chirakijja et al., 2024; Cohen et al., 2025). This suggests that a broader set of domestic appliances may have underappreciated health and welfare benefits.

In that regard, only a handful of economic studies have examined health-relevant nutritional indicators such as vitamin and mineral intakes (Berry et al., 2020; von Grafenstein et al., 2023; Wong et al., 2014). Our findings highlight that economic research can uncover the nutritional impacts of non-health interventions, thereby broadening the scope of economic research on nutrition and human capital development.

Finally, this paper contributes methodologically by introducing a within-adopter design based on appliance functionality that can be replicated in standard household surveys to study the welfare effects of durable goods when randomization is infeasible. Prior to our study, evidence on refrigeration was largely descriptive. Most work compared households with and without refrigerators and no study effectively addressed selection bias (Heard et al., 2020; Karlsson et al., 2020; Martinez et al., 2021; Jo et al., 2024; Karlsson and Subramanian, 2023; Visram and Brown, 2020).[3] By exploiting the GLSS measure of

[2]Prior work reinforces this perspective. Lee et al. (2016) show that home solar systems cannot substitute for grid power in rural Kenya, primarily because they cannot support high-wattage appliances such as refrigerators. Follow-up work by the same authors (Lee et al., 2020a) shows that households willing to pay for connections benefit more than those who connect only when subsidized, highlighting the importance of complementary inputs. More recent work (Burlig and Preonas, 2024) finds that the impacts of electrification depend on factors such as community size and service demand, pointing to the importance of how electricity is used, rather than simply whether it is available.

[3]Several studies were conducted outside Africa, where dietary patterns and energy infrastructure differ considerably (Craig et al., 2004; Heard et al., 2020; Karlsson et al., 2020). Moreover, some contributions

broken versus functioning refrigerators and purchase information, we implement a within-adopter design that can be replicated to study other technologies such as air conditioners and water pumps. Random allocation of appliances is rarely feasible, but nationally representative surveys can record ownership and functionality at relatively low cost, which could enable similar analyses of the effect of appliance take-up.

In section II, we provide relevant context about food insecurity and refrigerator ownership in Ghana. Section III presents the food insecurity data, the estimation method and the corresponding results. Section IV presents the food basket data and the corresponding impacts of refrigeration on basket composition. Section V presents additional data on nutrient intakes and the corresponding impact of refrigeration on nutrition. Section VI concludes.

II Context: Diet and Refrigeration in Ghana

Food insecurity, whether moderate or severe, remains a critical global challenge. In low- and middle-income countries, vulnerabilities in food storage infrastructure, especially under warmer climates, increase the risk of spoilage and constrain access to perishable foods.

Downstream, food waste from Ghanaian households, at 84 kg per person per year, is estimated to exceed the global average of 79 kg UNEP (2024), despite having lower-than-average food access.[4]

Better cold chains could make a significant difference (Friedman-Heiman and Miller, 2024). Refrigeration extends the shelf life of perishable foods and enhances food safety (Karlsson and Subramanian, 2023; Martinez et al., 2021; Faber et al., 2009; Ballantine

focused narrowly on children under five, potentially underestimating broader population-level effects (Grantham-McGregor et al., 2007; Karlsson and Subramanian, 2023; Martinez et al., 2021; Steyn et al., 2006). In addition, even research on food waste, where refrigeration might be expected to play a central role, has overlooked the role of cooling technologies: a systematic review of 54 interventions targeting food waste did not include any analysis of household refrigeration (Liechti et al., 2024).

[4]Post-harvest losses alone are estimated to be about 30% in Ghana (Ehrlich, 2025; Ghanaian Times, 2022) despite efforts from the National Food Buffer Stock Company (NAFCO), which manages surplus cereals during glut and lean periods (Afriyie et al., 2023; Armah et al., 2019). Nutrient-rich foods, such as fruits, vegetables, eggs, and dairy, are often handled by individual consumers; however, no national policy currently actively addresses household-level food waste. In fact, such consumer-focused policies remain largely absent across many developing countries (Aulakh et al., 2013; Majumder et al., 2016).

et al., 2008). At the household level, it also enables bulk purchases with lower unit costs, while reducing transportation expenses and providing time savings (Karlsson and Subramanian, 2023; Martinez et al., 2021; Craig et al., 2004).

In Ghana, the gains from refrigeration are now within reach because the national electrification rate has doubled since 2000, reaching 88.5% in 2024. Improved electricity access facilitates appliance adoption, with refrigerators being one of the first significant investments for many families after electricity access (Lee et al., 2016; McNeil and Letschert, 2010). Between 1993 and 2022, the proportion of Ghanaian households owning functioning refrigerators rose markedly, from 1.2% to 22.4% in rural areas and from 23.4% to 56.3% in urban areas. Similar trends can be observed in many LMICs where refrigeration is currently being adopted at pace (ICF, 2022).[5]

Nonetheless, one-fifth of the world's population still lacks a refrigerator (IEA, 2022). Households without refrigerators sometimes rent freezer spaces. The cost reported by Afriyie et al. (2023) in their community study is 2 to 5 Ghanaian Cedis[6] per day, which is equivalent to 13% to 34% of Ghana's 2023 daily minimum wage of 14.88 Cedis. This economic burden excludes many households from the benefits of refrigeration.

As for those who own a refrigerator, not all appliances are of the same quality. Ghana (and for that matter Africa) has persistently been a dumping ground for inefficient and obsolete refrigerators imported from other regions, such as Europe, China or North America. Aside from being near or over their life-usage capacity, these imported obsolete appliances are often not designed for tropical climates and are therefore prone to frequent breakdowns, reducing their useful life upon arrival in Ghana (Tamakloe, 2022; Agyarko et al., 2021).[7]

When refrigerators break down, repairs are often costly, and there are few qualified

[5]However, economic disparities have remained a barrier to refrigeration adoption. The ownership rate of functioning refrigerators among the wealthiest households (fifth quintile of income) increased from 43.1% in 1993 to 93.8% in 2022, but ownership rates remain minimal among poor households (0.7% and 8.2% for the first and second quintiles, respectively) (ICF, 2022, 1993). Other factors, like frequent power outages and unstable electricity supply, also affect appliance functionality and sustainability. Illegal or substandard electrical connections further exacerbate these issues, increasing the risk of appliance damage.

[6]This is equivalent to $0.34 to $0.86 at the time of their study, using the Bank of Ghana exchange rate (Bank of Ghana, 2024a).

[7]In 2000, more than 200,000 obsolete fridges were imported to Ghana, and over 400,000 second-hand refrigerators were imported into Ghana between 2008 and 2012. In 2014 alone, about 9,500 of such refrigerators were illegally imported (Tamakloe, 2022; Agyarko et al., 2021).

technicians available to perform them (FAO-UN, 2016). Many households hold on to broken refrigerators for extended periods, hoping to repair them in the future, since the appliance typically represents a significant investment. Many households also retain non-functional refrigerators even when they have no intention of repairing them, using them for non-cooling storage, for instance (Obeng-Odoom and Amedzro, 2011).[8]

Ghanaian households without adequate refrigeration typically rely on unconventional food preservation methods (Afriyie et al., 2023, 2022). These include placing half-used canned milk in cold water, mixing onions with lime, applying wood ash to sliced yams, and adding charcoal to soups before storage. Smoking is another common practice for preserving fish, meats, poultry, and snails. While these methods are culturally rooted, they often result in food poisoning and can pose significant health risks.[9]

Adequate food storage could increase resilience to food shocks, particularly where diets are fragile. Annan et al. (2022) found that most Ghanaian households consume unbalanced diets with adverse nutritional outcomes. Coomson and Aryeetey (2022) found a high prevalence of underweight and wasting among young children, who often suffer from anaemia and vitamin A deficiency.[10]

Ensuring access to affordable, functional refrigeration, while strengthening cold chain infrastructure and addressing appliance quality and disposal, could play a key role in improving both food security and nutrition outcomes in Ghana and similar contexts.

[8]The disposal of large electronic appliances is rare. While the country prohibits improper disposal of bulky electronic waste, it lacks accessible and complementary systems for proper disposal or recycling (FAO-UN, 2016). Both economic and practical constraints (particularly the cost and difficulty of moving heavy, bulky appliances) significantly limit households' options for responsible disposal. Instead, broken refrigerators are frequently repurposed for non-perishable storage, thus preserving some utility even in a non-functional state (Obeng-Odoom and Amedzro, 2011). In lower-income households, a broken refrigerator may also serve as a source of spare parts for other appliances. Moreover, refrigerator ownership is sometimes viewed as a status symbol in certain communities; displaying a refrigerator, whether functional or not, can convey social standing and pride in possession. Some families may also attach sentimental value to their refrigerators (especially when they were gifts or inherited items), which further discourages disposal, even when the appliance is broken (Obeng-Odoom and Amedzro, 2011).

[9]For instance, high levels of toxic polycyclic aromatic hydrocarbons (PAHs) have been detected in traditionally smoked fish (Moorthy et al., 2015). PAHs are toxic and carcinogenic. Among other health risks, chronic exposure is associated with lung cancer (Moorthy et al., 2015). Besides, high salt concentrations in preserved meats can increase hypertension risk (Dzikunoo et al., 2021; Plahar and Lu, 1989).

[10]They also found that 6.8% of children under 5 suffer from wasting, and 17.5% from stunting, and almost half of all deaths in children under 5 are attributable to malnutrition. Meanwhile, unhealthy weight gain is also on the rise. Rousham et al. (2020), focusing on Ghana and Uganda, show a consistently lower consumption of fruits and vegetables but a high consumption of sugar-sweetened beverages, which contribute to overweight and obesity. This issue is now affecting adult women with increasing concern, with overweight rates rising from 13% in 1993 to 43% in 2022 (ICF, 1993, 2022).

III Impact of refrigeration on perceived food insecurity

Data

This paper uses the last two waves of the Ghana Living Standards Surveys (GLSS VI and VII) (GSS, 2025).[11] Those surveys provide information from 16,772 households in 2012-2013 and 14,004 households in 2016-2017, respectively. We extract several variables on appliance ownership, food insecurity, and food expenditure. We also extract several socioeconomic and demographic characteristics. Since we are interested in households with a refrigerator, the study focuses on households with access to reliable electricity.[12]

Depending on data availability for specific outcomes, analyses either use both waves or GLSS VII only, which is more comprehensive. For food insecurity, we use GLSS VII only, which is when the GSS adopted the Food Insecurity Experience Scale (FIES) developed by the FAO. The scale covers 8 binary questions that require respondents to self-report on their ability to access food of sufficient quality and quantity in the last 12 months. For instance, the questions include whether the household was worried about not having enough food to eat, or whether the respondent did not eat for a whole day. Each response is assigned the value of 1 if the household reports experiencing food insecurity (and zero, if otherwise).

With these 8 questions, we create two main food insecurity indicators. The first is a food insecurity risk indicator which takes the value of 1 if a household responds "yes" to at least 1 of the eight questions (and zero, if otherwise). The second is a count measure

[11] The Ghana Statistical Service (GSS) used a two-stage stratified sampling design. In the first stage, a nationally representative sample consisting of 1000-1200 enumeration areas was selected from all over the country to form the primary sampling units (PSUs), which were allocated into the 10 administrative regions of the country using a probability proportional to population size (PPS). At the time of both surveys, there were 10 administrative regions in Ghana. The PSUs were divided into rural and urban areas, followed by a complete listing of all households in the identified PSUs. This formed the secondary sampling units (SSUs). At the second stage, 15 households were systematically selected from each PSU for the survey. This yielded a nationwide sample size of 18,000 households in GLSS VI and 15,000 households in GLSS VII. The response rates for GLSS VI and VII were 93.2% and 93.3%, respectively. Each survey was spread over 12 months in order to ensure a continuous recording of household consumption and expenditures and changes occurring thereof.

[12] Households that depend solely on crop residue, firewood, flashlights, and candles for electricity power were excluded (that is, 3% of the households owning fridges), leaving only those who depend on the national grid, mini-grid, and or solar power.

indicating the intensity of food insecurity experienced by the household. It is obtained from summing up all 8 indicators; hence, it ranges from 0 to 8, and higher values represent higher food insecurity intensity.

Table 1 provides summary statistics on the food insecurity responses, separately for households with a functioning or a broken refrigerator. On average, households with broken fridges are at much higher risk of food insecurity and are more likely to answer yes to several food insecurity questions.

The differences reported in **Table** 1 are not causal due to substantial differences between both groups. For instance, households with broken fridges bought lower-quality appliances. In addition to appliance ownership and functionality, the GLSS includes the purchase price and the year of acquisition of appliances.[13] For some households, refrigerators were received as gifts and, therefore, a price was not reported. **Table** 2 provides summary statistics for these main variables, separately for functioning and broken refrigerators. To reduce risks of measurement error and recall bias, we have winsorized the 1st and 99th percentiles of the price and age of refrigerators. Broken refrigerators were usually older and bought at lower prices.[14] Those offered as gifts were also more likely to break down.

The GLSS include many variables related to household characteristics, such as household size or income, as well as the educational attainment or marital status of the household members. **Tables** B1 and B2 in Appendix B show that the sociodemographic characteristics of those with broken refrigerators also differ. They are predominantly male-headed, with larger household sizes, lower formal education, and lower income.

Our modelling systematically selects relevant sociodemographic and socioeconomic variables as control variables through a standard LASSO process. In short, LASSO is a common machine learning technique that allows refining the pool of potential covariates and identifying the variables that have a significant correlation with the probability of being treated. We used this process to select potentially important additional sociodemo-

[13]The prices of the appliances for a few households were reported in foreign currencies (USD, Euros, Pounds, CFA, Naira, etc.). We converted all prices to Ghanaian Cedis (GHS) using historical end-of-year average exchange rates from FRED (2024), the Bank of Ghana (2024b), and the World Currency Exchange Rates and Currency Exchange Rate History (2024). We furthermore converted all prices to 2017 real prices, adjusting for inflation using Ghana's 2017 CPI deflator provided by the World Bank (2025).

[14]In 2017, GHS 1 = \$0.23 on average (Bank of Ghana, 2024b).

Table 1: **Mean statistics for food insecurity indicators**

Food insecurity outcomes	(1) **Functioning fridges** (N = 3,388)	(2) **Broken fridges** (N = 252)	(3) **Diff.**	(4) **SE**
Panel A: Constructed indices				
A1: At risk of food insecurity (0/1)	0.37	0.65	-0.27***	0.03
A2: Intensity of food insecurity (0 to 8)	1.52	3.01	-1.49***	0.16
Panel B: Disaggregated indices				
B1: Worried about not having enough food to eat	0.29	0.52	-0.23***	0.03
B2: Unable to eat healthy and nutritious food	0.23	0.44	-0.21***	0.03
B3: Ate only a few kinds of foods	0.27	0.53	-0.27***	0.03
B4: Had to skip a meal	0.21	0.42	-0.21***	0.03
B5: Ate less than expected	0.22	0.47	-0.26***	0.03
B6: Ran out of food	0.17	0.35	-0.18***	0.03
B7: Hungry but did not eat	0.11	0.20	-0.09***	0.02
B8: Did not eat for a whole day	0.03	0.07	-0.04***	0.01

Notes: Data from GLSS VII only. N = total number of households in the sample category. SE = Standard Error of the mean difference. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 2: **Mean statistics for refrigerators**

Refrigerator information	(1) **Functioning fridges** (N = 7,639)	(2) **Broken fridges** (N = 490)	(3) **Diff.**	(4) **SE**
Purchase price (2017 GHS)	800	658	+142***	35
Age (in years)	4.59	6.87	-2.28***	0.16
Gift (=1)	0.07	0.14	-0.07***	0.01

Notes: Data from GLSS VI and GLSS VII, pooled. N = total number of households in the sample category. Prices are deflated to 2017 real local currency values. SE = Standard Error of the mean difference. *** p<0.01, ** p<0.05, * p<0.1.

graphic and socioeconomic variables after pre-selecting primary determinants (e.g., the price of the refrigerators). The LASSO process ensures parsimony and avoids a "*throw in the kitchen sink*" or any arbitrary, ad-hoc approach in covariate selection, which could bias estimates (Shortreed and Ertefaie, 2017).

In using LASSO, researchers pre-select the minimum set of essential variables and then let LASSO select any additional ones from a large pool of potential covariates. In

addition to fridge characteristics in **Table** 2, we pre-selected information on household location (region and urban/rural dummies), household size, and the age and gender of the household head. We also control for the year and month of interviews. This allows for temporal, geographical, and compositional balance. We then let LASSO select any additional relevant variable from the available set of potential covariates. In all cases, the LASSO model selected household income as the only additional control variable. **Table** B1 in **Appendix** B presents summary statistics for the variables selected by LASSO. **Table** B2 provides information on the full set of covariates available for selection.

Estimation

Our estimation of the impact of refrigeration on food insecurity is a LASSO-based Augmented Inverse Probability Weighting (AIPW) estimator, which is among the set of machine learning estimators for causal inference described in Chernozhukov et al. (2018). This combines the LASSO selection of control variables with an estimation of treatment effects, similar in spirit to a matching estimator comparing a treatment and a control group after controlling for observables. This estimation approach allows us to compare the food insecurity of similar households, living in the same region and locality (urban/rural), who bought their refrigerators at the same time and for similar prices.

The treatment group consists of households that reported that their refrigerator was broken at the time of the interview, and the control group comprises those with a functioning refrigerator. This choice of treatment and control groups is an important methodological contribution to the literature, which, so far, has compared households with and without refrigerators.[15]

We use the food insecurity variables described above as outcome variables. Separately for each outcome variable, we estimate the average treatment effect on the treated (ATT) of having a broken fridge at the time of the interview. The ATT therefore captures a disruption in access to refrigeration. Some access may persist, for example if households rent refrigerated storage, which is costly, or rely on family support by using another

[15]In **Appendix** F, we show that our choice to focus on people with broken vs. functioning appliances increases the comparability of the control and treatment groups. With our data, results looking at the composition of food baskets while comparing households with and without refrigerators would not be robust to hidden bias.

household's appliance. In routine circumstances, however, households must store food without access to modern refrigeration.

The core assumption of the model is that, after conditioning on observed covariates, the treatment and control groups are comparable. This requires that the odds of belonging to the treatment or control groups are similar across observations, which, in our case, is equivalent to ensuring that the refrigerators in both groups would have had the same probability of having broken down at the time of the interview. Then household assignment to the treatment or control group would be as good as random.

To ensure comparability, we use all the covariates in **Tables** 2 and B1. This includes, first and foremost, information on the price of purchase, the year of acquisition of refrigerators and whether the appliance was a gift. These variables constitute very good proxies for appliance quality.

We jointly perform the LASSO selection and the estimation of treatment effects. These two stages must be jointly estimated, as separate estimation would distort the standard errors, inflate type I errors, and potentially bias the estimates (StataCorp, 2021; Leeb and Pötscher, 2005, 2006, 2008). Following the recommendations of Belloni et al. (2014), we use Stata's `telasso` command (StataCorp, 2023; Koch et al., 2017; Shortreed and Ertefaie, 2017; Hastie et al., 2015; Tibshirani, 1996), which is specifically designed to estimate treatment effects using AIPW while incorporating LASSO-based covariate selection. `telasso` is doubly robust (i.e. either the outcome model or the treatment model can be misspecified) and Neyman orthogonal (i.e. robust to model selection errors).[16]

Finally, standard errors were clustered at the Enumeration Area (EA) level to adjust for intra-cluster correlation due to the two-stage sampling design of the surveys (as described in footnote 11).

[16]See **Appendix** B for more discussions on LASSO and `telasso`. `telasso` implements an AIPW estimator, using LASSO to select appropriate covariates from a pool of potential covariates. By jointly estimating variable selection and treatment effects, telasso ensures consistent causal inference. Furthermore, it is more resilient to minor machine learning errors than traditional matching techniques, such as regression adjustment (RA), inverse probability weighting (IPW), or combined methods like IPWRA. To select the optimal value of the LASSO penalty parameter, we employed the cross-validation (CV) method in all our baseline estimations. We use the `telasso` command in Stata 17 with the default setup and a random-number seed of 777 for reproducibility.

Food insecurity results

We begin by assessing standardized bias between treated and control groups before and after conditioning on covariates. **Figure** 1 presents standardized mean differences (SMD), variance ratios, and cumulative probability graphs.[17] Standard benchmarks (Rosenbaum and Rubin, 1985) recommend mean differences below 20% and variance ratios between 0.5 and 2, ideally close to 1. Therefore, our sample appears well-balanced with SMD close to zero and variance ratios close to 1. The cumulative probability graphs further confirm good overlap and balance in propensity scores within the common support region after conditioning. Overall, **Figure** 1 suggests that the treated and control groups are comparable.

In **Table** 3, we move on to showing our baseline treatment effects of broken refrigerators on food insecurity. Each row corresponds to a different food insecurity outcome and, therefore, a different estimation. Column (1) shows the ATT from the `telasso` estimator. The positive signs imply that having a broken fridge worsens food insecurity. Thus, on average, we find that households with a broken fridge are 16 percentage points more likely to be at risk of food insecurity (see **Panel A**, outcome A1). Column (2) displays the treated group mean for comparison. In relative terms, losing access to refrigeration increases the risk of food insecurity by 33 percent relative to what the risk would have been if the fridge were functioning.[18]

We also find that households without a fridge are more likely to answer yes to several questions: having a broken fridge raises the intensity of food insecurity by 0.92 points, which is about a 44% increase relative to what the intensity would have been had the fridge been functioning (outcome A2).

In **Panel B**, disaggregated results by question are statistically significant for 7 out of 8 questions. Refrigeration appears to reduce households' concerns about whether they will have enough food and might help them avoid running out of food. In addition, refrigeration seems to support greater dietary diversity: households with functioning refrigerators are more likely to eat healthy and nutritious foods and less likely to rely on

[17] Given that we have 10 food insecurity outcomes, it would be ideal to present 10 covariate balance graphs. However, since the same covariates were selected in all models and the figures were exactly the same, we report only one figure to save space. All the other figures can be made available upon request.

[18] This is: $\frac{\text{ATT}}{(\text{treated mean - ATT})} = \frac{0.16}{0.65-0.16} \approx 0.33$

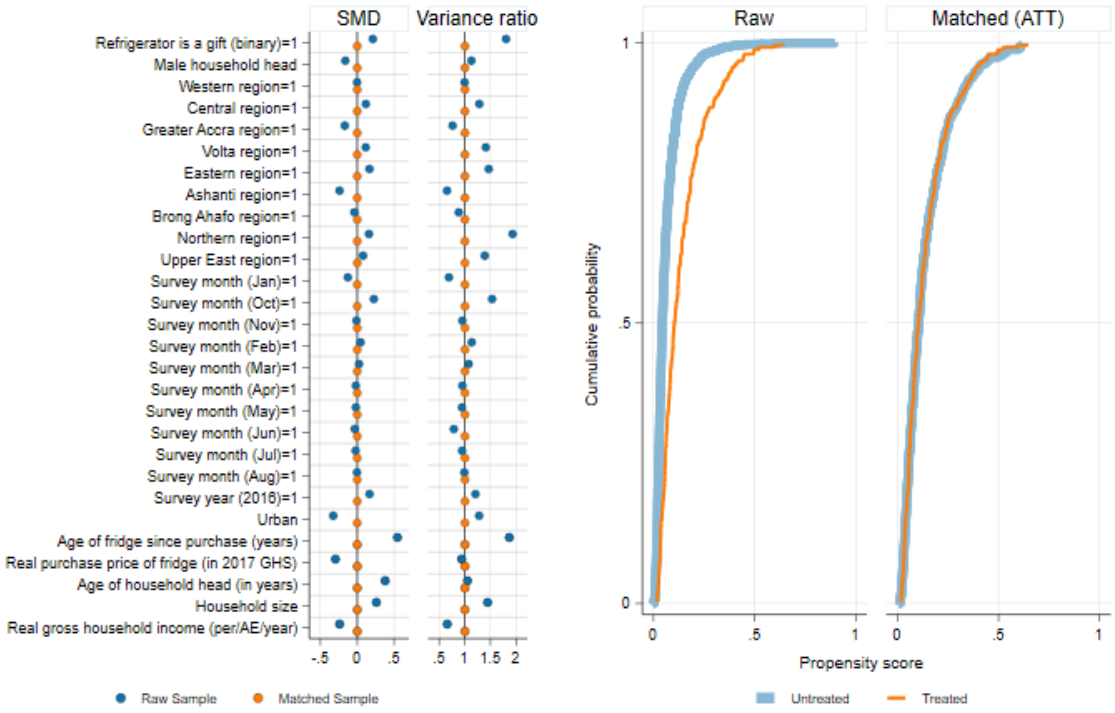


Figure 1: Model diagnostics for the food insecurity outcomes

Notes: Diagnostic tests when running our model on food insecurity outcomes, using GLSS VII data only. The left panel presents the covariate balance using the standardized mean difference (SMD) and variance ratio. In the raw sample, we observe large disparities, even beyond 20% in the SMD and outside the recommended range in the variance ratio. However, in the matched sample and under the common support, the SMD is zero (0) and the variance ratio is one (1) for all covariates. The right panel presents the cumulative probability plots of the propensity scores before matching (Raw) and after matching (Matched) under the common support. ATT = Average Treated effect on the Treated. The vertical axis of the right panel reports the probability that the score is less than or equal to a given propensity score.

a limited variety of foods. These households may also skip meals less frequently. Finally, functioning refrigerators might help some households avoid running out of food, going without food for an entire day, or being hungry but unable to eat.

We check whether all these results are robust to hidden bias in Column (3) by reporting the results of Rosenbaum bounds tests. As already explained, the core assumption of the model is that the control and treatment groups are virtually identical after conditioning on covariates. Nonetheless, in practice, controls are often imperfect, and we could expect that households in both groups might not have had exactly the same probability of belonging to the treatment and control groups due to unobserved confounders such as household preferences and habits affecting appliance life spans (or simply because of slight differences in propensity scores). Differences in probabilities of treatment due to

unobservables could create a hidden bias in the estimates (Caliendo and Kopeinig, 2008).

We use the bounds test developed by Rosenbaum (2002a,b) to assess the sensitivity of our estimates to possibly remaining differences between treatment and control groups. The test relies on a sensitivity parameter Γ, which denotes the difference in the probability of treatment between the observations in the treated group vs. the control group. The bounds test assesses whether the treatment effect would remain of the same sign even if the odds ratio of belonging to the treatment vs. the control group were as large as Γ. Results should ideally remain valid even for large values of Γ, as this would indicate that they are robust to the potential presence of unobserved confounding factors (Becker and Caliendo, 2007; Diprete and Gangl, 2004; Liu et al., 2013; Rosenbaum, 2002a,b).[19]

Column (3) of **Table** 3 shows the maximum Γ value from the sensitivity analysis, which represents the largest tolerable difference in the odds of treatment between the groups for the results to remain statistically significant at the 5% level.[20] A maximum Γ of 370% (as shown in **Table** 3) implies that the odds of belonging to one group would have to differ by a factor of 1 to 4.7 for the results to become statistically insignificant, confirming the robustness of our main result to hidden bias. Results for other entries are also generally robust to hidden bias with a maximum Γ above 40%. Disaggregated results by question remain robust for five out of the seven statistically significant items.

[19]To implement this bounds test, we re-estimate treatment effects with entropy balancing (Hainmueller, 2012), using the same pre-selected and LASSO-selected baseline covariates to ensure full comparability with the `telasso` estimates. This is because we cannot use `telasso` to formally perform sensitivity tests for hidden bias. We report in **Appendix** C1 the ATTs obtained from entropy balancing. They are very similar to the `telasso` estimates. Our entropy balancing estimations are done in STATA17 using the default settings of the `kmatch` command developed by Jann (2020). We set the entropy balance constraint to the 2nd-order moment. Other forms of matching algorithms are nearest neighbor, caliper, and radius, stratification and interval, and inverse probability weighting. Our choice presents two advantages considering our sample size and the number of LASSO-selected covariates to control for. Because kernel methods are non-parametric, they do not require specific functional form specifications and, therefore, are not subject to misspecification biases. Furthermore, control households with propensity scores that are closer to the scores of the treated households are assigned more weight than those with far-away scores. This allows us to reduce the biases that distant matches could create, considering that, with a relatively small sample size and a high number of controls, matches could be imperfect. Moreover, to match households on the selected covariates, we use a kernel method (Heckman et al., 1998). There are many kernel-based algorithms to choose from. We adopt an Epanechnikov kernel pair-matching algorithm since it is the most frequently used in the literature. The smoothing parameter is automatically determined through the pair-matching approach as discussed in Huber et al. (2013). Frölich (2004) provides details on other algorithms.

[20]For example, a Γ of 10% means that unobserved factors would need to increase the odds of being in the treatment group by more than 10% (relative to similar control households) in order to overturn our results. In this example, results would be barely robust to hidden bias since a 10% difference in odds is small.

Table 3: **Causal impacts of having a broken fridge on food insecurity**

Food insecurity outcomes	(1) **ATT**	(2) **Treatment group mean**	(3) **Max. Γ**
Panel A: Constructed indices			
A1: Risk of food insecurity	0.16***	0.65	370%
	(0.04)		
A2: Intensity of food insecurity	0.92***	3.01	40%
	(0.20)		
Panel B: Disaggregated indices			
B1: Worried about not having enough food to eat	0.14***	0.52	150%
	(0.04)		
B2: Unable to eat healthy and nutritious foods	0.13***	0.44	70%
	(0.04)		
B3: Ate only a few kinds of foods	0.16***	0.53	160%
	(0.04)		
B4: Had to skip a meal	0.15***	0.42	50%
	(0.04)		
B5: Ate less than the respondent's expectation	0.14***	0.47	90%
	(0.04)		
B6: Ran out of food	0.13***	0.35	<10%
	(0.04)		
B7: Went without eating for a whole day	0.05	0.20	<5%
	(0.03)		
B8: Hungry but did not eat	0.04***	0.07	<5%
	(0.02)		

Notes: N = 3,615. Standard errors are in parentheses. Panel A focuses on the 2 main indicators constructed from the data, while Panel B focuses on the 8 original indicators from GLSS VII. Column (1) reports the ATT; Column (2) reports the mean of the treated group; and Column (3) reports the maximum Γ value (in %) from the Rosenbaum bounds sensitivity analysis beyond which our estimates would be sensitive to hidden bias, at 5% critical level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Robustness

We first assess the stability of the AIPW estimates (in **Table** C2 of **Appendix** C). Results are similar when adding additional household controls (education and employment status of the household head; materials used for the roof, walls, and floor of the dwelling; and ownership of any other broken appliance) and when excluding households that received their refrigerator as a gift. Estimates are also unchanged when using the adaptive LASSO rather than the cross-validation method for covariate selection. Replacing AIPW with inverse probability weighting, propensity score matching, or Mahalanobis distance matching yields similar effects. Finally, the GLSS includes survey weights, which we have not used in the main estimations since our sample, by construction, cannot be represen-

tative of the Ghanaian population. We use those weights as importance weights in **Table** C2, which has no substantial effect on our findings. Taken together, these checks show that the main findings do not depend on estimator choice or covariate specification.

We then assess whether omitted variable bias could be driving the results. Already, the high maximum Γ values of the Rosenbaum bounds tests in **Table** 3 give us strong confidence in the direction of the effects. Furthermore, since we control for appliance price and time of purchase, and households have little control over whether a refrigerator of a given age and price breaks down, the remaining heterogeneity due to unobserved factors, such as household "*savviness*" or maintenance behaviour, is likely to play only a small role. Matching and AIPW estimators are appropriate when treatment assignment is exogenous (i.e., appliance breakdown is not a choice) and the covariate set is rich enough to balance treated and control groups. However, we cannot fully rule out the existence of unobserved factors that may influence both refrigerator durability and food insecurity. Results could therefore still be biased even if the critical value for Γ is high (Becker and Caliendo, 2007). A high Γ only gives us strong evidence that the main results are robust to hidden bias, even though bounding tests cannot directly confirm the unconfoundedness assumption.

To evaluate potential selection concerns, we run two complementary analysis that rule out two potential sources of omitted variable bias: first using the sample of refrigerators received as a gift, and second with placebo tests focusing on other broken appliances.

Sample of households with refrigerators as gift

We estimate fixed-effects models using only households that received a refrigerator as a gift. This "*gift*" sample is less prone to selection bias as appliance ownership in this group is only indirectly related to household characteristics or preferences. Comparability is further enhanced since all households received their appliance from an external source. We run linear regressions weighted by the survey weights to assess the impact of refrigerator breakdowns on food insecurity within this subsample.

Table 4 presents the results for the "*gift*" sample. The estimates are remarkably consistent with those from the main quasi-experimental design. In both samples, refrig-

erator breakdowns are associated with higher food insecurity by about 0.15 points for the risk score and 0.8–0.9 points for the intensity score. The magnitude of the coefficients closely matches the baseline estimates despite the smaller sample size and wider confidence intervals, suggesting that the main results are not driven by appliance selection or unobserved heterogeneity.

Table 4 furthermore reports six specifications used to test the stability of results to the inclusion of additional control variables. If appliance breakdowns are exogenous for households that received a refrigerator as a gift, the coefficients should remain stable despite the inclusion of controls. Results are stable across specifications despite a loss of precision due to small sample sizes and the inclusion of many controls, providing further evidence that our results are not driven by omitted variable biases.

Table 4: **Impact of refrigerator breakdowns on food insecurity among households that received the appliance as a gift**

	(1)	(2)	(3)	(4)	(5)	(6)
A. Food security indices						
A1: At risk of food insecurity (0/1)	0.153 (0.108)	0.166 (0.111)	0.163 (0.125)	0.135 (0.118)	0.117 (0.120)	0.087 (0.124)
A2: Intensity of food insecurity (0 to 8)	0.786* (0.471)	0.871* (0.475)	0.889* (0.470)	0.735 (0.464)	0.704 (0.464)	0.644 (0.477)
B. Disaggregated indicators						
B1: Worried about not having enough food	0.132 (0.096)	0.151 (0.097)	0.160 (0.104)	0.131 (0.102)	0.129 (0.099)	0.131 (0.111)
B2: Unable to eat nutritious foods	0.135 (0.093)	0.149 (0.094)	0.160* (0.091)	0.149 (0.093)	0.160* (0.096)	0.177* (0.103)
B3: Ate only a few kinds of foods	0.205** (0.102)	0.206** (0.102)	0.232** (0.109)	0.177* (0.097)	0.164* (0.090)	0.119 (0.091)
B4: Had to skip a meal	0.153 (0.096)	0.182* (0.097)	0.161* (0.093)	0.127 (0.084)	0.107 (0.082)	0.082 (0.091)
B5: Ate less than the respondent's expectation	0.096 (0.087)	0.085 (0.086)	0.081 (0.079)	0.051 (0.075)	0.059 (0.079)	0.015 (0.092)
B6: Ran out of food	0.012 (0.063)	0.026 (0.062)	0.019 (0.062)	0.001 (0.064)	0.010 (0.065)	0.039 (0.081)
B7: Went without eating for a whole day	0.013 (0.058)	0.032 (0.061)	0.027 (0.059)	0.040 (0.065)	0.010 (0.069)	0.024 (0.078)
B8: Hungry but did not eat	0.040 (0.052)	0.040 (0.052)	0.048 (0.051)	0.057 (0.053)	0.066 (0.049)	0.057 (0.043)
Controls						
Age of appliance		✓	✓	✓	✓	✓
Region			✓	✓	✓	✓
Rural/urban				✓	✓	✓
Household var.					✓	✓
Community var.						✓
Observations	268	268	267	267	267	198

Notes: All specifications include interview year and interview month fixed effects. Observations include only households that were given the refrigerator in GLSS VII. Household variables include (i) age and (ii) sex of householder; (iii) whether ever attended school; (iv) their employment status (employed or not); (v) the household size; and whether the (vi) walls, (vii) roof and (viii) walls of the dwelling are permeable. Community variables are not available for all observations and include: access to a market, a financial institution, a road, living conditions of the area, and the community's main activity (fishing, trading, handicraft, other). Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Placebo tests

Refrigerator breakdowns may proxy for household characteristics such as care, "*savviness*," or maintenance ability, in which case food insecurity would respond to appliance failures generally rather than to the loss of refrigeration specifically. Breakdown rates could also reflect unobserved home conditions, such as indoor heat or humidity. Finally, appliance breakdowns could influence spending patterns if households save to replace appliances, which could act as a negative income shock. To assess the role of these mechanisms, we estimate placebo effects using the same identification strategy as in the main analysis but applied to other household appliances.

The GLSS records functional status, purchase year, and price for 14 additional appliance categories (see **Appendix** A). However, many of these categories contain too few broken units to support estimation. The two appliances with sufficient variation in functionality in GLSS VII are radios and sewing machines. We therefore construct placebo treatments that consist of having a broken radio or a broken sewing machine, with the corresponding control groups defined as households with functional units of the same appliance.

Table 5 displays the placebo results. We find no effects of these placebo treatments on food insecurity or dietary composition. This indicates that the estimated impacts are specific to refrigeration rather than to general appliance failure, unobserved household characteristics linked to failures, or income-related replacement behaviour.

Overall, we find that refrigeration significantly reduces food insecurity. It alleviates concerns about food shortages and supports greater food diversification. Covariate balance diagnostics confirm that the treated and control groups are well-matched. The results are also robust to hidden bias, changes in control variables, and alternative estimation strategies. Placebo tests further emphasize that the observed effects are specific to refrigeration.

Table 5: **Placebo tests - Impacts of other broken appliances on food insecurity**

Food insecurity outcomes	**Broken radio (N=1,467)**	**Broken sewing machine (N=588)**
Panel A: Constructed indices		
A1: Risk of food insecurity	0.04	-0.04
	(0.07)	(0.09)
A2: Intensity of food insecurity	0.21	-0.28
	(0.36)	(0.49)
Panel B: Disaggregated indices		
B1: Worried about not having enough food to eat	0.05	-0.03
	(0.07)	(0.10)
B2: Unable to eat healthy and nutritious foods	0.04	-0.12
	(0.07)	(0.08)
B3: Ate only a few kinds of foods	0.09	-0.12
	(0.07)	(0.09)
B4: Had to skip a meal	0.05	-0.06
	(0.06)	(0.09)
B5: Ate less than the respondent's expectation	0.04	0.01
	(0.07)	(0.09)
B6: Ran out of food	-0.02	-0.02
	(0.06)	(0.07)
B7: Went without eating for a whole day	-0.03	0.05
	(0.05)	(0.08)
B8: Hungry but did not eat	0.0006	0.02
	(0.02)	(0.04)

Notes: N = number of observations in each sample category. Columns (1) and (2) report the ATT of having a broken radio and sewing machine on food insecurity, respectively. Standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

IV Impact of refrigeration on food baskets

We next examine whether refrigeration fosters food diversity. We focus on acquired quantities, which we derive from purchases and own consumption by the household. We report technical details on how we compute food expenditure and food quantities in **Appendix** D.

In short, food expenditure data is available in GLSS VI and GLSS VII. Enumerators visited each household seven times at five-day intervals. During these visits, a diary was used to record daily consumption and market value of own produce, as well as food expenditures. On the first visit, a literate household member was identified and trained to record all household expenditures daily, submitting the diary during the next visit for data entry. If no literate person was present in the household, the enumerator would make daily visits to record expenditures. The daily values were then aggregated. Both

methods led to the computation of 30-day expenditure by food item for each household after either seven visits at 5-day intervals, or daily visits for 30 days in the case of illiterate households.

We compute quantities by dividing the value of own consumption and food expenditures by local market prices. To collect information for computing food and non-food CPIs, the Ghana Statistical Service also required enumerators to visit three randomly selected local markets in each community. GLSS VI and VII therefore professionally collected consistent price information from local markets for each food item. For every food item, enumerators either observed or weighed the item using standard scales and recorded both the quantity and the corresponding retail price. Reported prices are highly reliable: in over 98% of cases, the three item-specific recorded prices are similar within a community.

Summary statistics

Summary statistics for food quantities are reported in **Table** 6, distinguishing households with functioning versus broken refrigerators. Total food acquisition amounts to about 1.6 kg per person per day, roughly half of which consists of vegetables (high in water content). The distribution across food groups is also consistent with expectations: Ghanaian households rely primarily on vegetables and cereals, complemented by smaller amounts of meat, dairy, eggs, and fish. Sweetened beverages are rarely purchased, consistent with evidence that they are treated as an occasional luxury rather than a daily staple.[21] On average, households with a broken refrigerator acquire less of meat, dairy, eggs and fish. They also acquire more of roots, tubers and plantain, and less of leafy vegetables and fruit. Finally, they acquire non-alcoholic beverages (e.g. sweetened drinks) less often. This is consistent with households being predominantly of lower income status. Summary statistics for food expenditure are provided in **Appendix section** D.3.[22]

[21] Observed acquired quantities are close to daily recommendations. For animal-sourced products, the National Food-Based Dietary Guidelines by Ghana's Ministry of Food and Agriculture (2023) recommend 0.144kg per day, which is higher than our sample averages (of 0.181kg and 0.132g for people with a functioning or broken refrigerator, respectively). Quantities for leafy vegetables also align with recommendations of 0.227kg per day, but households may not eat enough fruit, and pulse, nuts and seeds, with the recommendation being twice higher for fruit, at 0.226kg per day, and four times higher for pulse, nuts and seeds, at 0.198kg per day.

[22] With the GSS, we can track food expenditure outside the home, such as in canteens, restaurants, and hotels. However, we are unable to disaggregate such expenditure into item-specific values or to estimate

Table 6: **Summary statistics of food quantities**

Food group (in kg/day/AE)	**(1)** **Functioning fridges (N = 7,639)**	**(2)** **Broken fridges (N = 490)**	**(3)** **Diff.**	**(4)** **SE**
Total at home (excl. alcohol)	1.659	1.576	0.083	0.086
All Meats	0.044	0.030	0.015***	0.003
Milk & dairy products	0.015	0.009	0.005***	0.001
Eggs	0.038	0.021	0.018***	0.004
Fish	0.084	0.072	0.013***	0.005
All vegetables	0.783	0.827	-0.044	0.053
Roots, tubers, & plantain	0.530	0.615	-0.085**	0.042
Other leafy vegetables	0.228	0.202	0.026*	0.015
Fruits	0.127	0.103	0.024***	0.009
Cereals, flour & bread	0.290	0.298	-0.009	0.018
Pulse, nuts, & seeds	0.056	0.057	0.000	0.005
Oil & fats	0.038	0.040	-0.001	0.003
Spices	0.017	0.016	0.001	0.002
Sugar & sweets	0.018	0.018	0.000	0.002
Alcoholic beverages	0.015	0.013	0.003	0.003
Non-alcoholic beverages	0.042	0.027	0.015***	0.003

Notes: Data from GLSS VI and GLSS VII, pooled. N = total number of households in the sample category. SE = Standard Error of the mean difference. *** p<0.01, ** p<0.05, * p<0.1.

Impact of refrigeration on food quantities

We estimate average treatment effects on the treated for food quantities using the same LASSO-based AIPW estimator (Chernozhukov et al., 2018), as for food insecurity in the previous section. The baseline model is strictly identical, and results are reported in **Table** 7.

We find sizeable statistically significant reductions in animal-sourced foods. The effects are statistically significant at the 1% level for all meats, milk and dairy products, and eggs, and at the 10% level for fish. In relative terms, having a broken refrigerator reduces meat acquisition by about 21%, milk and dairy by 17%, eggs by 39%, and possibly fish by 10% relative to how much would have been demanded if the fridge were functioning.

Column (3) further shows that reductions in animal-sourced foods are robust to hidden

their calorie contents (see **Appendix** D).

bias. Model diagnostics, robustness checks, and placebo tests are reported in **Appendix section** D.2. These mirror those for food insecurity: results are consistent and stable across specifications, and the effect is specific to refrigeration and not to other appliances. In the sample of refrigerators received as a gift, we again find similar effects for animal-sourced products, with no meaningful changes for non-animal foods. Furthermore, we report ATTs for food expenditure instead of food quantities in **Appendix section** D.3, confirming that refrigeration has an impact on access to animal-based food items.

The results in this section confirm our prior results for food insecurity, which suggested that refrigeration may allow for higher food diversity. They are also consistent with the fact that refrigeration can durably extend the shelf life of perishable products such as meat and dairy, which is particularly important in a tropical country with hot weather like Ghana (Karlsson and Subramanian, 2023; Martinez et al., 2021; Faber et al., 2009; Ballantine et al., 2008). Furthermore, we find no effect on the products that rarely go to the refrigerator, such as pulses or cereals.

V Impact on nutritional availability

We compute the caloric and nutrient content of all food quantities using the West Africa Food Composition Table by Vincent et al. (2020). This database reports macronutrient and micronutrient values per 100 grams of the edible portion of each food item. We merge this item-level nutritional database with our food quantity data to compute the nutritional composition of foods acquired, including caloric content, carbohydrates, proteins, fats, vitamins, and minerals. The data can be further disaggregated to analyse the origin of nutrients (e.g., animal-sourced versus plant-based) or specific nutrient types such as fat subcategories, vitamins, or minerals.

This procedure allows us to estimate nutritional availability at the point of acquisition. Our approach represents a significant contribution to the literature: to our knowledge, no previous study has estimated the causal impact of household refrigeration on nutrition using item-level quantity-to-nutrient conversion. More generally, only a handful of economic studies examine detailed impacts on micronutrients (Berry et al., 2020; von Grafenstein et al., 2023; Wong et al., 2014), even though deficiencies are widespread in

Table 7: **Impacts of broken fridges on food quantities**

Food groups	**(1) ATT**	**(2) Treatment group mean**	**(3) Max. Γ**
Total at home (excl. alcohol)	-0.052	1.576	70%
	(0.096)		
All Meats	-0.008***	0.030	160%
	(0.003)		
Milk & dairy products	-0.002***	0.009	110%
	(0.001)		
Eggs	-0.013***	0.021	390%
	(0.003)		
Fish	-0.008*	0.072	80%
	(0.005)		
All vegetables	-0.010	0.827	70%
	(0.062)		
Roots, tubers, & plantain	0.014	0.615	<5%
	(0.052)		
Other leafy vegetables	-0.010	0.202	120%
	(0.013)		
Fruits	-0.003	0.103	80%
	(0.009)		
Cereals, flour & bread	0.010	0.298	<5%
	(0.020)		
Pulse, nuts, & seeds	0.004	0.057	<5%
	(0.006)		
Oil & fats	0.003	0.040	<5%
	(0.003)		
Spices	0.001	0.016	<5%
	(0.001)		
Sugar & sweets	0.002	0.018	<5%
	(0.002)		
Alcoholic beverages	0.001	0.013	<5%
	(0.002)		
Non-alcoholic beverages	-0.003	0.027	140%
	(0.003)		

Notes: N=8,072. Standard errors are in parentheses. Column (1) reports the ATT, Column (2) reports the mean of the treatment group; Column (3) reports the maximum Γ value (in %) from the Rosenbaum bounds sensitivity analysis beyond which our estimates would be sensitive to hidden bias, at 5% critical level. *** p<0.01, ** p<0.05, * p<0.1.

low-income countries and can lead to severe health consequences.

Table 8 reports the average per-capita, adult-equivalent daily intakes computed using this method. Both total caloric intake and the distribution across carbohydrates, fats, and proteins are consistent with physiological norms and with existing evidence for Ghana (Ministry of Food and Agriculture, 2023; UNDP, 2013). On average, animal-sourced

foods account for about 11% of total caloric intake but provide roughly half of protein intake, approximately 20% of fat intake, 16% of mineral intake, and about 7% of vitamin intake. This aligns with the composition of typical Ghanaian dishes, which are primarily based on cereals, pulses, nuts and seeds, and vegetables (especially roots, tubers, and plantain).[23]

Importantly, these summary statistics do not account for food waste at the household level, which was not recorded in GLSS and is notoriously difficult to measure.[24] Because food waste is likely lower among households with refrigeration, our estimated treatment effects on food intake may understate the true nutritional benefits of refrigeration. The reported estimates should therefore be interpreted as conservative. In future versions of this paper, we will provide alternative estimates under the assumption of 10% food waste for households without refrigeration and 5% for those with refrigeration, to illustrate how differential waste levels could affect our findings. These values of 5 and 10 percent are consistent with the available evidence on food waste and preservation potential in SSA.[25]

Another limitation is that the analysis captures average intakes and does not observe short-run fluctuations. Refrigeration may enable more regular, smaller intakes of perishable foods, which could improve micronutrient absorption and reduce temporary privation relative to infrequent, one-off consumption. As a result, our estimates may understate the health benefits of refrigeration as a consumption-smoothing technology.

[23] In **Appendix** E, we compare summary statistics of average intakes between the treated and control groups, distinguishing animal sources (columns 1–4) from plant sources (columns 5–8). We find significant differences in nutrient intakes from animal origin, and some differences too for a few plant-based nutrients (iron, zinc, fibre and some vitamins).

[24] Existing studies either rely on self-reported waste (which tends to underestimate true levels because households are reluctant to report wasting food) or on weighing household waste (UNEP, 2024). They are field studies which are limited in scope (e.g. to a few households). In practice, because organic waste is often mixed with other materials, and because some discarded items (such as bones or vegetable peels) are inedible in any case, accurate estimates of edible food are difficult to gather.

[25] Existing research suggests that household food waste in SSA is relatively low. Estimates place it at around 10% of total food expenditure in Nigeria (Sunday et al., 2022). Higher-value foods such as meat are typically wasted less often than cheaper, more perishable items. For Ghana, UNEP (2024) estimates per-capita household food waste at 84 kg per year, including non-edible components. In our dataset, the average annualized per-capita food acquisition is 481 kg, implying that total food waste (including non-edible portions) represents roughly 18% of total acquisition. This estimate is broadly consistent with an edible waste share of about 10% as found in Nigeria. Households frequently report that limited storage contributes to spoilage, suggesting that refrigeration reduces food waste (e.g., Sunday et al. (2022)). Empirical evidence from China supports this view: households with refrigerators waste less food after controlling for demographic characteristics (Zhang et al., 2025).

Table 8: **Mean caloric intake corresponding to the quantities of food acquired**

Nutrient intake (unit/day/AE)	(1) Animal origin	(2) Plant based	(3) Total
Metabolizable energy (kcal)	277	2321	2599
Carbohydrate (g)	3	392	394
Protein (g)	36	39	76
Fat (g)	14	63	76
Saturated fat (g)	5.1	25.3	30.4
Monounsaturated fat (g)	5.1	14.2	19.7
Polyunsaturated fat (g)	1.6	11.4	13.0
Cholesterol (g)	0.29	-	0.29
Minerals (g)	1.7	8.9	10.6
Iron (mg)	3.5	24.4	27.9
Zinc (mg)	3.2	12.6	15.6
Calcium (mg)	350	324	674
Phosphorous (g)	0.57	0.96	1.53
Potassium (g)	0.50	3.55	4.05
Magnesium (mg)	54.5	400	455
Copper (mg)	0.19	2.68	2.88
Vitamins (mg)	17.76	256	273
Vitamin A (RAE) (mg)	0.08	0.93	1.02
Retinol (mcg)	82.6	-	82.6
Beta-carotene equivalents (provitamin A) (mg)	-	10.96	10.96
Alpha-carotene (provitamin A) (mg)	-	0.86	0.86
Beta-carotene (provitamin A) (mg)	-	10.03	10.03
Vitamin B1 (mg)	162	1682	1844
Vitamin B2 (mg)	0.27	2.02	2.28
Vitamin B3 (mg)	13.3	17.3	30.6
Vitamin B6 (mg)	0.49	2.34	2.83
Vitamin B9 (mg)	0.03	0.38	0.41
Vitamin B12 (mcg)	7.99	-	7.99
Vitamin C (mg)	0.5	223	223.5
Vitamin D (mcg)	4.94	0.68	5.62
Vitamin E (mg)	2.89	8.31	11.2
Fibre (g)	-	53.1	53.1

Notes: N=8,129. Animal origin comprises all meat, eggs, milk and dairy products, and fish. Plant-based refers to the main remaining food groups, as described in the data section. Vitamin A (RAE) is the retinol activity equivalent of vitamin A. It comprises the fractional sum of beta-carotene equivalent and retinol, whereas beta-carotene equivalent is the fractional sum of beta-carotene, alpha-carotene, and beta-cryptoxanthin. Hence, total Vitamins is the sum of Vitamin A (RAE), Vitamin B1, Vitamin B2, Vitamin B3, Vitamin B6, Vitamin B9, Vitamin B12, Vitamin C, Vitamin D, & Vitamin E. See Vincent et al. (2020) for more details on the definition of each nutrient.

Nutritional impact of refrigeration

For now, **Table** 9 reports the average treatment effect on the treated (ATT) of refrigeration on nutritional availability, estimated with the LASSO-based AIPW estimator described above. We also report Rosenbaum bounds tests to assess the sensitivity of our estimates to hidden bias.

Results for main nutrients are disaggregated by source (animal vs. plant-based) and then aggregated to compute overall ATTs. We find that households with broken refrigerators obtain significantly fewer nutrients from animal-sourced foods. The overall reduction of 47 calories corresponds to a 19-percent decline in animal protein intake, and this estimate is robust to hidden bias.[26] By contrast, we find no statistically significant effect of losing access to refrigeration on plant-based nutritional values. Point estimates are positive and suggest that households without refrigeration may partly offset losses in animal-sourced foods by consuming easier-to-store, plant-based alternatives. Taken together, the results indicate no statistically significant effect on total nutritional intake once animal- and plant-based sources are summed, consistent with the possibility that households adapt by substituting toward plant-based options when refrigeration is unavailable.

In **Table** 9, we aggregated vitamins and minerals. Results for each individual vitamin and mineral are provided in **Appendix** E. For vitamins that are available from both animal and plant sources, we find no statistically significant differences in total intake once both sources are combined. However, some micronutrients are only available in animal-sourced products. This is the case for cholesterol, vitamin B12 and retinol, whose results are reported in **Table** 10.

The added cholesterol for those with functioning refrigerators has, most likely, no implications for human health. This is because saturated fats, which are associated with an increase in low-density lipoprotein in blood (bad cholesterol levels) do not increase (see **Table** 9). The lower retinol and vitamin B12 intakes reported in **Table** 10 are particularly concerning for households with a broken refrigerator, especially given that this group includes a large share of food-insecure households likely to experience vitamin

[26] The mean of the treatment group is 195 kcal/day/AE. The impact of refrigeration loss is equal to: -47/[196 - (-47)] $\approx$ 19%.

Table 9: **ATT of broken fridge on home-based nutrient intake (excl. alcohol)**

Nutrient intake	**Animal origin**		**Plant based**		**Total (excl. alcohol)**	
	ATT	**Max. Γ**	**ATT**	**Max. Γ**	**ATT**	**Max. Γ**
Metabolizable energy (kcal)	-47.1***	110%	140.0	<5%	94.46	<5%
	(12.6)		(124.2)		(130.13)	
Carbohydrate (g)	-0.37***	110%	23.23	<5%	22.85	<5%
	(0.14)		(22.47)		(22.50)	
Protein (g)	-5.37***	100%	3.36	<5%	-2.218	40%
	(1.66)		(2.46)		(3.40)	
Fat (g)	-2.68***	160%	4.29	<5%	1.61	<5%
	(0.69)		(3.62)		(3.95)	
Saturated (g)	-0.99***	150%	1.35	<5%	0.28	<5%
	(0.25)		(1.72)		(1.84)	
Monounsaturated (g)	-1.02***	170%	1.51	<5%	0.45	<5%
	(0.28)		(0.96)		(1.10)	
Polyunsaturated (g)	-0.30***	140%	0.95	<5%	0.65	<5%
	(0.09)		(0.73)		(0.75)	
Minerals (g)	-0.28***	110%	0.35	<5%	0.09	<5%
	(0.08)		(0.46)		(0.50)	
Vitamins (mg)	-2.51***	110%	9.497	<5%	6.94	<5%
	(0.79)		(16.512)		(16.87)	
Fibre (g)	-	-	4.00	<5%	4.00	<5%
	-		(3.54)		(3.54)	

Notes: N=8,072. Standard errors are in parentheses.

deficiencies.

Vitamin deficiency in SSA is difficult to quantify because it requires collecting biological samples. Deficiencies were, however, often observed at high prevalence, especially for vitamin B12 (e.g., Adu-Afarwuah et al. (2025); Sakyi et al. (2021)). Using quantitative dietary intake data of young children in rural Northern Ghana, De Jager et al. (2018) estimated that food intakes were likely insufficient to cover vitamin A and vitamin B12. A similar analysis of the diets of 400 pregnant women in Ghana found that over 50 percent obtained less than two-thirds of the recommended daily amounts of vitamin B12 (Saaka and Oladele, 2020). For individuals experiencing such deficiencies, refrigeration could have positive health impacts that further field research could directly quantify.

Vitamin A can be obtained directly from retinol (active vitamin A) or synthesized by the body from plant-based provitamin A. Active vitamin A not only provides direct absorption but also enhances the conversion of provitamin A, playing a dual protective

role against vitamin A deficiency. The health consequences of vitamin A deficiency can be severe, including an increased risk of mortality (e.g., Baye et al. (2022)). Vitamin A deficiency impairs immune function, increases susceptibility to infections, and is a leading cause of preventable blindness, particularly among children.

Vitamin B12, by contrast, can only be obtained through the consumption of animal products. Deficiency is associated with megaloblastic anaemia, impaired neurological development, and cognitive decline. Among women of reproductive age, low vitamin B12 status has been linked to adverse pregnancy outcomes and an increased risk of neural tube defects in infants (e.g., Dhume et al. (2023)), particularly given the higher vitamin B12 requirements during pregnancy.

Hence, animal-sourced micronutrients can play a crucial role in combating severe vitamin deficiencies. Additional health benefits from refrigeration may also arise from increased mineral bioavailability. Even if total mineral intake remains stable, as shown in **Table** 9, minerals are not absorbed equally by the human body. For instance, heme iron from meat is absorbed far more efficiently than non-heme iron from plant sources, and meat consumption also enhances the absorption of non-heme iron. Similarly, zinc and calcium from animal-sourced foods are generally more bioavailable than their plant-based equivalents.

Table 10: **Impact of refrigeration on nutrients strictly available from animal-sourced foods**

Nutrient intake (unit/day/AE)	**ATT**	**Max. Γ**
Retinol	-16.49***	120%
	(4.35)	
Vitamin B12	-1.47***	110%
	(0.58)	
Cholesterol	-0.06***	150%
	(0.02)	

Notes: N=8,072. Retinol and Vitamin B12 are measured in micrograms/day/AE. Cholesterol is measured in grams/day/AE. Standard errors are in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Below, we report exploratory findings for the ATT of broken refrigerators on iron intake, accounting for bioavailability. In future research, we intend to leverage data from micronutrient surveys in Ghana and other African countries to determine whether refrigeration could have a causal impact on ferritin concentrations in blood samples. For

now, we follow the reference values of Carpenter and Mahoney (1992) to distinguish absorption rates from animal and plant sources. These values vary with physiological iron stores (typically low, at around 0–250 mg, among anaemic individuals) and with the dietary availability of iron, which increases with meat intake but tends to be low in plant-based diets. For individuals with near-zero iron stores and low dietary availability, reference absorption rates are 35% for heme iron and 5% for non-heme iron.

Table 11: **Impact of broken fridge on bioavailability-corrected estimates for iron**

Physiological iron stores	**Dietary availability**	**Total**	**Heme iron**	**Non-heme iron**
0 – 250mg	Low	-9.76	-19.60***	9.99
		(10.77)	(6.12)	(7.02)
	Medium	2.14	-19.60***	21.97
		(18.43)	(6.12)	(15.44)
250mg – 500mg	Low	-7.81	-15.68***	7.99
		(8.62)	(4.90)	(5.61)
	Medium	-1.79	-15.68***	13.98
		(12.38)	(4.90)	(9.82)
500mg – 1000mg	Low	-6.95	-12.88***	5.99
		(6.74)	(4.02)	(4.21)
	Medium	-2.98	-12.88***	9.99
		(9.20)	(4.02)	(7.02)
Above 1000mg	Low	-4.47	-8.40***	3.99
		(4.45)	(2.62)	(2.81)
	Medium	-2.47	-8.40***	5.99
		(5.67)	(2.62)	(4.21)

Notes: Bioavailability reference rates were taken from Carpenter & Mahoney (1992). Standard errors are in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

As shown in **Table** 11, results suggest a potential reduction in total iron absorption for anaemic individuals with low iron stores and low dietary availability, although confidence intervals remain wide. These estimates are conservative, as they imperfectly account for the fact that heme iron enhances dietary availability for all iron, especially non-heme iron. With the data at hand, considerable uncertainty remains, as reductions in heme iron intake may be offset by increases in non-heme iron. Moreover, substantial heterogeneity across individuals is plausible, given that bioavailability depends on numerous factors, including enhancers and inhibitors of iron absorption, individual nutritional requirements

(which are substantially higher for pregnant women, as discussed earlier), and variation in iron content and bioavailability across vegetables and meats. For instance, if some households substitute meat with plant-based foods that are low in iron or inhibit its absorption, refrigeration could, in some cases, protect against anaemia.

VI Conclusion

With rapid electrification across LMICs, an estimated 800 million people are expected to gain access to a refrigerator in the next decade (IEA, 2022). The consequences could be profound for the global fight against hunger.

This paper provides the first causal evaluation of refrigeration's effects on food insecurity, diets and welfare using quasi-experimental variation in appliance functionality at the time of interview. Conditioning on location, purchase price, acquisition year, and other observable determinants of breakdown, we compare otherwise similar Ghanaian households with functional versus broken refrigerators, estimating causal effects with a machine learning algorithm (i.e. LASSO-based AIPW design).

We find that refrigeration can provide serious health benefits by reducing food insecurity and enhancing diets. First, access to a functional refrigerator reduces household food insecurity risk by 33 percent and make households less likely to report food-insecurity experiences (e.g., running out of food, skipping meals, or reducing meal size) and show lower intensity on summary indices. These effects are robust to hidden bias, consistent across several alternate specifications and specific to refrigerators (using placebos). Second, refrigeration shapes the composition of food demand: households with broken fridges demand 10-40% less of animal-sourced items (meat, dairy, eggs, fish). Third, the primary nutritional margin could be quality rather than quantity: loss of refrigeration is associated with a 19-percent decline in access to animal-source protein. While total calories remain stable, households with lost access to refrigeration are at higher risk of vitamin B12 deficiency. Vitamin A absorption may also be impaired without refrigeration, which enhances retinol intake. Mineral deficiencies, such as iron deficiency, might also concern more Ghanaian households without access to refrigeration, even though confidence intervals are large and iron absorption is context dependent.

Overall, our results highlight that distribution and storage frictions limiting access to essential goods can lead to substantial welfare losses. Refrigeration is an overlooked but essential channel through which households can become more resilient to shocks to food security, and a key mechanism linking electrification to improved welfare. Future research could analyse impacts on health markers to better capture nutritional and health outcomes at a micro level. The environmental implications of the above will also require special scrutiny, since refrigeration can be expected to trigger higher meat consumption and higher energy usage. Likewise, the diffusion of appliances should probably be accompanied with information policies, for instance, on how to best use refrigeration to enhance diets. Finally, our identification strategy could be used to study other technological transitions in low- and middle-income countries.

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Appendices

A Summary statistics for other household appliances

Table A1: **Mean statistics of appliance ownership (GLSS VI & VII pooled sample)**

Appliance information	**Broken**		**Functioning**			
	N (1)	**Mean** (2)	**N** (3)	**Mean** (4)	**Diff.** (5)	**SE** (6)
Real price of radio	139	106	3486	122	15.5	20.1
Age of radio	139	4.98	3486	3.58	-1.4^{***}	0.28
Real price of fan	142	174	6963	172	-1.9	16.7
Age of fan	142	5.22	6963	3.98	-1.2^{***}	0.28
Real price of sewing machine	112	232	1493	415	183^{*}	105
Age of sewing machine	112	15.9	1493	11.5	-4.42^{***}	0.92
Real price of electric iron	92	103	6707	111	7.80	12.6
Age of electric iron	92	4.52	6707	3.23	-1.3^{***}	0.22
Real price of blender	96	158	2371	240	81.6^{**}	31.3
Age of blender	96	3.47	2371	2.95	-0.52^{**}	0.21
Real price of television	102	396	7602	827	431^{***}	105
Age of television	102	5.95	7602	4.49	-1.5^{***}	0.26
Real price of mobile phone	36	164	7778	431	267^{**}	115
Age of mobile phone	36	2.28	7778	1.88	-0.40	0.25
Real price of gas stove	63	361	4864	410	48.4	82.4
Age of gas stove	63	5.68	4864	4.15	-1.54^{***}	0.38
Real price of kerosene stove	33	431	133	314	-118	301
Age of kerosene stove	33	9.42	133	8.35	-1.07	1.51
Real price of electric stove	9	344	149	1296	951	1360
Age of electric stove	9	5.11	149	5.13	0.02	1.64
Real price of freezer	39	732	827	1937	1205^{**}	449
Age of freezer	39	7.67	827	4.83	-2.84^{***}	0.68
Real price of air conditioner	7	1003	202	1681	678	626
Age of air conditioner	7	5.29	202	4.16	-1.13	1.78
Real price of washing machine	14	2084	188	1652	-432	710
Age of washing machine	14	5.50	188	3.78	-1.72^{*}	0.88
Real price of microwave	31	372	975	433	60.71	88.7
Age of microwave	31	3.71	975	3.08	-0.63^{*}	0.36

Notes: Data from GLSS VI & GLSS VII, pooled. N = number of households owning each asset with non-missing information on the price and age of the appliance. All prices are deflated to 2017 real GHS, and all ages are in years. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table A2: **Mean statistics of appliance ownership (only GLSS VII sample)**

Appliance information	**Broken**		**Functioning**			
	N **(1)**	**Mean** **(2)**	**N** **(3)**	**Mean** **(4)**	**Diff.** **(5)**	**SE** **(6)**
Real price of radio	64	128	1406	134	6.28	28.9
Age of radio	64	5.67	1406	3.72	-1.95^{***}	0.40
Real price of sewing machine	88	387	3383	1016	629^{***}	163
Age of sewing machine	88	6.63	3383	4.99	-1.6^{***}	0.38
Real price of fan	58	187	3168	191	3.78	25.9
Age of fan	58	5.60	3168	4.33	-1.3^{***}	0.43
Real price of electric iron	57	116	2977	121	4.69	17.6
Age of electric iron	57	5.04	2977	3.59	-1.4^{***}	0.36
Real price of blender	42	211	1024	270	58.90	50.2
Age of blender	42	4.02	1024	3.28	-0.74^{**}	0.33
Real price of gas stove	37	345	2205	463	117	110
Age of gas stove	37	5.95	2205	4.54	-1.41^{**}	0.48
Real price of television	35	134	553	474	340^{*}	185
Age of television	35	17.8	553	11.3	-6.5^{***}	1.61
Real price of mobile phone	3	280	3556	514	235	422
Age of mobile phone	3	2.33	3556	2.17	−0.17	0.82
Real price of kerosene stove	5	79.4	19	50.1	−29.3	38.4
Age of kerosene stove	5	16.8	19	15.37	−1.43	5.94
Real price of electric stove	1	598	45	1277	679	.
Age of electric stove	1	7.00	45	5.60	−1.40	.
Real price of freezer	15	616	306	2584	1967^{***}	843
Age of freezer	15	7.60	306	5.21	-2.39^{**}	1.18
Real price of air conditioner	5	1152	93	1972	820	939
Age of air conditioner	5	5.40	93	5.00	−0.51	2.56
Real price of washing machine	7	415	99	1895	1480	954
Age of washing machine	7	5.43	99	4.10	−1.35	1.12
Real price of microwave	6	328	372	502	174	201
Age of microwave	6	6.50	372	3.62	-2.88^{***}	0.79

Notes: Data from GLSS VII only. N = number of households owning each asset with non-missing information on the price and age of the appliance. All prices are deflated to 2017 real GHS, and all ages are in years. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

B Technical description of LASSO and the covariate selection process

To strengthen identification, we incorporate additional socioeconomic controls. Given the large set of potential variables and to avoid arbitrariness in selection, we use a LASSO-based procedure rather than relying solely on a theory-driven approach.

LASSO for covariate selection

LASSO refines a large set of candidate predictors by shrinking coefficients through a penalization parameter (λ). It minimizes the residual sum of squares subject to this penalty, where $\lambda \geq 0$ governs the strength of regularization. Larger values of λ induce greater sparsity by shrinking more coefficients toward zero, potentially excluding relevant predictors, whereas very small values of λ imply little shrinkage.

Because the choice of λ critically affects model performance (Schneider and Wagner, 2012), it must be selected objectively. Common approaches include information criteria such as cross-validation (CV), the Akaike Information Criterion (AIC), and the Bayesian Information Criterion (BIC) (Tibshirani and Taylor, 2012). In small samples, AIC and BIC can be computationally cumbersome and more volatile than CV (Zou et al., 2007), whereas CV directly minimizes out-of-sample prediction error.[27] In all baseline specifications, we therefore adopt 10-fold cross-validation, which is the default in Stata 17.

Tables B1 and **??** report the full set of covariates used in the analysis. Table B1 presents the pre-selected controls together with the only variable selected by LASSO, income, which is therefore included in the baseline specification. Table **??** lists the remaining candidate covariates that were considered but not selected by LASSO.[28]

[27] More detailed discussions of LASSO are provided by StataCorp (2023), Shortreed and Ertefaie (2017), Hastie et al. (2015), and Tibshirani (1996).

[28] The variables in Table **??** also constitute a subset of the sociodemographic variables available in the survey. They include variables that could, in principle, affect selection into treatment, based on economic evidence, Ghana's cultural context, and related empirical studies (Agedew et al., 2023; Heard et al., 2020; Katoch, 2022; Martinez et al., 2021). To avoid scale effects and unintended penalization in the LASSO regression, all continuous covariates are standardized to have zero mean and unit standard deviation.

Use of `telasso` rather than LASSO

Some studies apply LASSO as a first step and then conduct causal inference using a separate model, effectively ignoring the uncertainty introduced by control selection or implicitly assuming that the selected model is correctly specified (StataCorp, 2023).

Ignoring model selection can inflate type I error rates, bias estimated treatment effects, and understate standard errors (Leeb and Pötscher, 2005, 2006, 2008; **?**). To address these concerns, Belloni et al. (2014) introduced methods that integrate LASSO-based variable selection with valid post-selection inference, allowing for causal interpretation.

`Telasso` is a machine-learning implementation of an Augmented Inverse Probability Weighting estimator. It simultaneously estimates treatment effects while using LASSO to select relevant covariates from a pre-specified pool, in addition to the pre-selected controls. Its main advantages are: (i) joint covariate selection and causal effect estimation, (ii) double robustness, (iii) Neyman orthogonality, and (iv) greater resilience to model misspecification, as it is less sensitive to machine-learning imperfections than single-equation estimators such as regression adjustment (RA), inverse probability weighting (IPW), or IPW with regression adjustment (IPWRA). This integrated approach allows us to exploit high-dimensional controls while maintaining valid inference for the average treatment effect on the treated.

Table B1: Summary statistics for baseline covariates

Baseline covariates	**(1)** Functioning refrigerator ($N = 7639$)	**(2)** Broken refrigerator ($N = 490$)	**(3)** Diff.	**(4)** SE
Male household head (= 1)	0.72	0.65	$+0.07^{***}$	0.02
Age of household head (years)	44.25	49.07	-4.82^{***}	0.64
Real gross household income (per AE/year)	11,221	7,917	$+3{,}304^{***}$	798
Household size	3.90	4.51	-0.61^{***}	0.11
Urban (= 1)	0.77	0.66	$+0.11^{***}$	0.02
Region				
Western	0.11	0.11	0.00	0.01
Central	0.10	0.10	0.00	0.01
Greater Accra	0.24	0.19	$+0.05^{**}$	0.02
Volta	0.06	0.11	-0.05^{***}	0.01
Eastern	0.10	0.12	-0.02	0.01
Ashanti	0.21	0.16	$+0.04^{**}$	0.02
Brong Ahafo	0.08	0.07	$+0.01$	0.01
Northern	0.03	0.06	-0.02^{**}	0.01
Upper East	0.04	0.06	-0.02^{**}	0.01
Upper West	0.03	0.02	$+0.01$	0.01
Survey year				
2012	0.12	0.11	$+0.01$	0.01
2013	0.44	0.37	$+0.07^{***}$	0.02
2016	0.10	0.15	-0.05^{***}	0.01
2017	0.35	0.37	-0.02	0.02
Survey month				
January	0.10	0.11	-0.01	0.01
February	0.10	0.10	0.00	0.01
March	0.10	0.10	-0.01	0.01
April	0.10	0.09	$+0.01$	0.01
May	0.09	0.09	$+0.01$	0.01
June	0.01	0.01	0.00	0.00
July	0.10	0.09	$+0.01$	0.01
August	0.09	0.08	$+0.01$	0.01
September	0.10	0.07	$+0.03^{**}$	0.01
October	0.11	0.18	-0.07^{***}	0.01
November	0.08	0.06	$+0.02$	0.01
December	0.03	0.03	0.00	0.01

Notes: Data from GLSS VI and GLSS VII, pooled. N denotes the number of households in each refrigerator status category. SE is the standard error of the mean difference. AE denotes adult equivalence. Income is measured in real 2017 Ghanaian Cedis and is winsorised at the 1st and 99th percentiles. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table B2: Summary statistics for candidate covariates not selected by LASSO

Potential covariates	(1) Functioning refrigerator ($N = 7639$)	(2) Broken refrigerator ($N = 490$)	(3) Diff.	(4) SE
Household head information				
Ever attended school (= 1)	0.92	0.88	0.04***	0.01
Employed (= 1)	0.87	0.83	0.04**	0.02
Marital status				
Never married before	0.13	0.05	0.08***	0.02
Married	0.69	0.69	0.01	0.02
Divorced/separated/widowed	0.18	0.27	−0.09***	0.02
Religion				
No religion	0.03	0.05	−0.02*	0.01
Christian	0.82	0.79	0.03	0.02
Islamic	0.14	0.16	−0.01	0.02
Other religion	0.01	0.00	0.00	0.00
Ethnicity				
Akan	0.57	0.53	0.04	0.02
Ga-Dangme	0.09	0.06	0.03**	0.01
Ewe	0.12	0.18	−0.05***	0.02
Guan	0.03	0.04	−0.01	0.01
Gurma	0.01	0.01	0.00	0.01
Mole-Dagbani	0.11	0.12	−0.02	0.01
Grusi	0.02	0.01	0.01	0.01
Mande	0.01	0.01	0.00	0.00
Other	0.04	0.03	0.01	0.01
Household-level information				
Owns other broken appliances (= 1)	0.07	0.28	−0.21***	0.01
Impermeable roof material (= 1)	0.97	0.96	0.00	0.01
Impermeable floor material (= 1)	0.98	0.98	0.01	0.01
Impermeable wall material (= 1)	0.88	0.80	0.09***	0.01
Dwelling ownership				
Owning	0.36	0.44	−0.07***	0.02
Renting	0.39	0.31	0.07***	0.02
Rent-free	0.25	0.25	0.00	0.02
Perching	0.00	0.00	0.00	0.00
Squatting	0.00	0.00	0.00	0.00

Notes: Notes: Data from GLSS VI and GLSS VII, pooled. N denotes the number of households in each refrigerator-status category. SE is the standard error of the mean difference. Ownership of other broken appliances is a binary indicator equal to one if the household owns at least one broken freezer, washing machine, air conditioner, television, radio, fan, sewing machine, or iron. Roof material is impermeable if it is metal sheet, slate/asbestos, cement/concrete, or roofing tiles. Floor material is impermeable if it is cement/concrete, stone, burnt brick, vinyl tiles, ceramic/porcelain/granite/marble tiles, or terrazzo. Wall material is impermeable if it is concrete, cement, brick, or stone. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

C Additional estimations for food insecurity

C.1 Estimates using entropy balance

Table C1: TELASSO vs. Entropy balance estimates from the food insecurity analyses

Food insecurity outcomes	(1) TELASSO	(2) Entropy balance
Panel A: Constructed indices		
A1: Risk of food insecurity	0.16***	0.18***
	(0.04)	(0.03)
A2: Intensity of food insecurity	0.92***	1.01***
	(0.20)	(0.22)
Panel B: Disaggregated indicators		
B1: Worried about not having enough food to eat	0.14***	0.15***
	(0.04)	(0.04)
B2: Unable to eat healthy and nutritious foods	0.13***	0.14***
	(0.04)	(0.04)
B3: Ate only a few kinds of foods	0.16***	0.18***
	(0.04)	(0.04)
B4: Had to skip a meal	0.15***	0.15***
	(0.04)	(0.04)
B5: Ate less than the respondent's expectation	0.14***	0.17***
	(0.04)	(0.04)
B6: Ran out of food	0.13***	0.12***
	(0.04)	(0.04)
B7: Went without eating for a whole day	0.05	0.06*
	(0.03)	(0.03)
B8: Hungry but did not eat	0.04***	0.03*
	(0.02)	(0.02)

Notes: N=3,615. Standard errors are in parentheses. Each cell is a separate estimation. Column (1) provides the ATT from TELASSO as found in the baseline. Column (2) provides the corresponding ATT from entropy balance estimation, which uses the same preselected and LASSO-selected covariates employed in the baseline. Standard errors are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

C.2 Robustness of AIPW estimation for food insecurity indicators

We assess the sensitivity of our baseline results to specification changes. Key sensitivity checks on the AIPW model are reported in Table C2. Each cell represents an estimate from a unique AIPW estimation, and each row is a separate food insecurity outcome. Results are robust to the selection of control variables. In Column (1), we pre-select additional covariates that were not LASSO-selected in the baseline model: the education and employment status of the household head, the materials used in making the roof, wall, and floor of the dwelling, and whether the household owns a broken appliance other than a refrigerator. We observe impacts similar to the baseline estimates. Results are not impacted by the context of refrigerator acquisition either. In column (2), we exclude households that reported having their refrigerators as gifts. Furthermore, results do not depend on the chosen estimator. In column (3), we estimate the baseline model using the adaptive selection method in the LASSO process instead of the cross-validation method. In columns (4) to (6), we provide results with alternative matching estimation techniques: inverse probability weighting (IPW), propensity score matching (PSM), and Mahalanobis distance (MD) matching. Finally, the GLSS includes survey weights, which we have not used in the main estimations since our sample, by construction, cannot be representative of the Ghanaian population. Furthermore, `telasso` treats survey weights as importance weights rather than frequency weights. Column (7) provides the results where we use the survey weights. Results are consistent with the baseline estimates.

Table C2: Robustness - Alternative specifications of the food insecurity models

Food insecurity outcomes	Extra covariates	Excluding gifts	Adaptive LASSO	IPW	PSM	MD	Survey weights
Panel A							
A1:	0.16***	0.16***	0.16***	0.19***	0.21***	0.25***	0.16***
	(0.04)	(0.03)	(0.04)	(0.03)	(0.04)	(0.03)	(0.04)
	[390%]	[390%]	[390%]	[390%]	[110%]	[230%]	[390%]
A2:	0.89***	0.91***	0.92***	1.10***	1.13***	1.38***	1.01***
	(0.20)	(0.21)	(0.20)	(0.22)	(0.22)	(0.21)	(0.23)
	[40%]	[50%]	[80%]	[80%]	[60%]	[110%]	[80%]
Panel B							
B1:	0.14***	0.14***	0.14***	0.17***	0.19***	0.22***	0.13***
	(0.04)	(0.04)	(0.04)	(0.04)	(0.04)	(0.03)	(0.05)
	[140%]	[150%]	[150%]	[150%]	[90%]	[130%]	[150%]
B2:	0.12***	0.11***	0.13***	0.16***	0.16***	0.19***	0.12***
	(0.04)	(0.04)	(0.04)	(0.04)	(0.04)	(0.03)	(0.05)
	[60%]	[70%]	[70%]	[70%]	[40%]	[60%]	[70%]
B3:	0.15***	0.16***	0.16***	0.20***	0.19***	0.25***	0.19***
	(0.04)	(0.04)	(0.04)	(0.04)	(0.04)	(0.03)	(0.05)
	[160%]	[170%]	[160%]	[160%]	[80%]	[140%]	[160%]
B4:	0.14***	0.14***	0.15***	0.16***	0.16***	0.20***	0.14***
	(0.04)	(0.04)	(0.04)	(0.04)	(0.04)	(0.04)	(0.05)
	[50%]	[60%]	[50%]	[50%]	[30%]	[50%]	[50%]
B5:	0.13***	0.14***	0.14***	0.18***	0.17***	0.23***	0.15***
	(0.04)	(0.04)	(0.04)	(0.04)	(0.04)	(0.03)	(0.04)
	[90%]	[100%]	[90%]	[90%]	[50%]	[80%]	[90%]
B6:	0.12***	0.13***	0.13***	0.13***	0.14***	0.17***	0.17***
	(0.04)	(0.04)	(0.04)	(0.04)	(0.04)	(0.04)	(0.04)
	[<10%]	[10%]	[<10%]	[<10%]	[<5%]	[<5%]	[<10%]
B7:	0.04	0.05	0.05	0.07*	0.07*	0.08***	0.05
	(0.04)	(0.04)	(0.03)	(0.03)	(0.03)	(0.03)	(0.04)
	[<5%]	[<5%]	[<5%]	[<5%]	[<5%]	[<5%]	[<5%]
B8:	0.04***	0.05***	0.04***	0.04***	0.05***	0.04***	0.05***
	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.01)	(0.02)
	[<5%]	[<5%]	[<5%]	[<5%]	[<5%]	[<5%]	[<5%]
N	3615	3354	3615	3640	3640	3640	3615

Notes: N denotes the number of observations. Standard errors are reported in parentheses and the maximum Γ from the Rosenbaum bounds sensitivity analysis is reported in square brackets. Food insecurity outcomes are defined as follows: A1 = risk of food insecurity; A2 = intensity of food insecurity; B1 = worried about not having enough food to eat; B2 = unable to eat healthy and nutritious foods; B3 = ate only a few kinds of foods; B4 = had to skip a meal; B5 = ate less than the respondent's expectation; B6 = ran out of food; B7 = went without eating for a whole day; and B8 = hungry but did not eat. Column (1) includes additional covariates plausibly selected by LASSO in the baseline. Column (2) excludes households that reported the refrigerator as a gift. Column (3) uses adaptive LASSO. Columns (4)–(6) use IPW, PSM, and MD estimators. Column (7) applies survey weights treated as importance weights. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

D Methods and complementary analyses of food baskets

D.1 Computing food expenditures and quantities

Food expenditure data.

Food expenditure data is available in GLSS VI and GLSS VII. Enumerators visited each household seven times at five-day intervals. During these visits, a diary was used to record daily consumption of own produce and expenditures. On the first visit, a literate household member was identified and trained to record all household expenditures daily, submitting the diary during the next visit for data entry. If no literate person was present in the household, the enumerator would make daily visits to record expenditures. The daily values were then aggregated. Both methods led to the computation of 30-day expenditure by food item for each household after either seven visits at 5-day intervals or daily visits for 30 days in the case of illiterate households.

Food items were classified in thirteen main categories: (1) All meats; (2) Fish; (3) Eggs; (4) Milk and dairy products; (5) Oil and fats; (6) Fruits; (7) All vegetables (including potatoes and other tubers); (8) Cereals, flour, and bread; (9) Pulses, nuts, and seeds; (10) Spices and condiments; (11) Sugars and sweets; (12) Non-alcoholic drinks; and (13) Alcoholic drinks. GLSS VII used a more comprehensive list of 305 food items, compared to 118 in GLSS VI. However, both nomenclatures are similar. Both waves covered the main food types, but GLSS VII provided more specific breakdowns. We use the 13 broad categories from both waves to assess overall food expenditure. We also compute total food expenditure, excluding alcoholic beverages. Furthermore, we disaggregate the all-vegetables category into two finer subcategories: roots, tubers and plantains vs. other vegetables.

The GSS recommends harmonizing the raw data on food expenditure to account for: (a) the consumption of own food production, which is very common in rural areas; (b) substantial local price differences; and (c) caloric needs based on household age and sex composition. Therefore, we harmonize the food expenditure data following the standard Survey-based Harmonized Indicator Program (SHIP) Manual of GSS (2018, 2014).

For own consumption, the surveys cover food items that can be classified following the same typologies as the expenditure section. Enumerators asked for quantities and their overall monetary value, asking farmers to provide the farmgate price of the units they consumed. A 30-day item-specific expenditure was computed by summing own consumption values and purchases for each household. To account for cost-of-living variations, we adjust those values based on the GSS regional cost-of-living food indices. To account for household size, sex and age compositions, we apply the equivalence scale used by GSS, which is derived from the 10th Edition of the National Research Council's Recommended Dietary Allowances (National Academy of Sciences - National Research Council, 1989). All values are deflated to 2017 real values using food deflators from the GSS (2018) and then expressed in daily values (i.e., divided by 30). This results in daily per capita adult equivalent food expenditure values, expressed in 2017 local currency.

Food quantities data.

We derive food quantity from the food expenditure data. Technically, GLSS VI reports quantities consumed from own production, but only expenditures for purchased items. In contrast, GLSS VII records both expenditures and quantities (including purchased quantities) at each household visit, along with units of measurement. However, only about 6% of these quantities were reported in standard units. This reflects local practices in Ghana, where food is often measured in traditional units such as *olonka* (a volume measure that varies across regions) or by containers such as cans, which also differ in weight or volume. Converting these non-standard units into consistent measures would require detailed region- and item-specific conversion factors for each traditional unit, which are not available.

Instead, we convert the 30-day item-specific expenditures into quantities using detailed market price information. To collect information for computing food and non-food CPIs, the Ghana Statistical Service requires enumerators to visit three randomly selected local markets in each community. Both surveys professionally collect consistent price information from local markets for each food item. For every food item listed in the expenditure section, enumerators either observe or weigh the item using standard scales and record both the quantity and the corresponding retail price. Reported prices

are highly reliable: in over 98% of cases, prices are consistent across the three markets for a given item within a community. In GLSS VI, all market prices were collected in kilograms or litres, while in GLSS VII, about 95% were expressed in standard units (e.g., kg, g, mg, litres). We converted all reported mass units into kilograms, and for items recorded in volume (e.g., litres or millilitres), we used food-specific density estimates from Charrondiere et al. (2012) to convert them into kilograms.

We then compute item-specific average local market prices per kilogram. When a specific food item lacked a recorded local market price, we imputed its price using the average per-kilogram price of the same item within the relevant rural/urban community (defined by region and rural/urban status). In the few cases where no such price was available, we used the average price for the broader food group (e.g., all cereals) within the same rural/urban community. Items labelled as "other foods" (e.g., other cereals, other fruits) were excluded from this procedure to avoid misclassification.

To proxy food quantities, we use the pre-harmonised 30-day item-specific expenditure data (i.e., before adjusting for cost of living and household composition). We divide those by the corresponding average local market price per kilogram per item. This method yields food quantities (in kilograms) for about 96% of all reported food items. We then express them in daily values and scale them to account for household composition by applying the adult-equivalence scale used by GSS (2018, 2014) and National Academy of Sciences - National Research Council (1989). Finally, we winsorize food quantities at the 1st and 99th percentiles to reduce the influence of outliers.

D.2 Food quantity: model diagnostics and complementary results

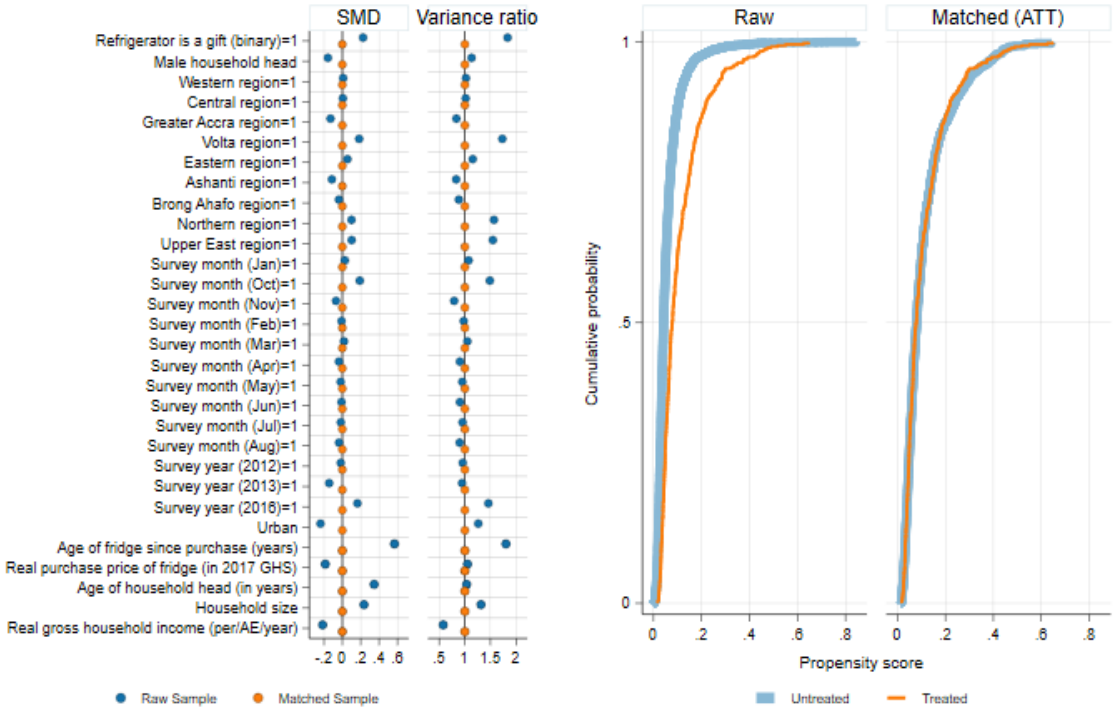


Figure D1: Model diagnostics for the food quantity outcomes

Notes: Diagnostic tests when running our model on food quantity outcomes, using pooled sample. The left panel presents the covariate balance using the standardized mean difference (SMD) and variance ratio. In the raw sample, we observe large disparities, even beyond 20% in the SMD and outside the recommended range in the variance ratio. However, in the matched sample and under the common support, the SMD is zero (0) and the variance ratio is one (1) for all covariates. The right panel presents the cumulative probability plots of the propensity scores before matching (Raw) and after matching (Matched) under the common support. ATT = Average Treated effect on the Treated. The vertical axis of the right panel reports the probability that the score is less than or equal to a given propensity score.

Table D1: TELASSO vs. Entropy balance estimates from the food quantity analyses

Food quantity category	(1) TELASSO	(2) Entropy balance
Total at home (excl. alcohol)	-0.052	-0.025
	(0.096)	(0.092)
All meats	-0.008***	-0.008***
	(0.003)	(0.002)
Milk & dairy products	-0.002***	-0.002***
	(0.001)	(0.001)
Eggs	-0.013***	-0.011***
	(0.003)	(0.003)
Fish	-0.008*	-0.005
	(0.005)	(0.005)
All vegetables	-0.010	0.009
	(0.062)	(0.060)
Roots, tubers, & plantain	0.014	0.041
	(0.052)	(0.049)
Other leafy vegetables	-0.010	-0.014
	(0.013)	(0.014)
Fruits	-0.003	0.004
	(0.009)	(0.009)
Cereals, flour & bread	0.010	0.009
	(0.020)	(0.021)
Pulse, nuts, & seeds	0.004	0.005
	(0.006)	(0.005)
Oil & fats	0.003	0.004
	(0.003)	(0.003)
Spices	0.001	0.001
	(0.001)	(0.002)
Sugar & sweets	0.002	0.002
	(0.002)	(0.002)
Alcoholic beverages	0.001	0.001
	(0.002)	(0.002)
Non-alcoholic beverages	-0.003	-0.002
	(0.003)	(0.003)
N	8072	8129

Notes: N = number of observations in the sample category. Each cell is a separate estimation. Column (1) reports the ATT from TELASSO as in the baseline specification. Column (2) reports the corresponding ATT from entropy balance estimation using the same preselected and LASSO-selected covariates. Standard errors are in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table D2: Alternative specifications for food quantities

Food quantity cateogry	(1) Extra covariates	(2) Excluding gifts	(3) Adaptive LASSO	(4) IPW	(5) PSM	(6) MD	(7) Survey weights
Total at home (excl. alcohol)	-0.055	-0.078	-0.052	-0.100	-0.108	-0.012	-0.086
	(0.095)	(0.088)	(0.096)	(0.105)	(0.110)	(0.089)	(0.103)
	[60%]	[70%]	[60%]	[70%]	[40%]	[40%]	[60%]
All meat	-0.008***	-0.006***	-0.008***	-0.008***	-0.009***	-0.010***	-0.008***
	(0.003)	(0.003)	(0.003)	(0.002)	(0.003)	(0.002)	(0.002)
	[140%]	[140%]	[160%]	[160%]	[110%]	[160%]	[140%]
Milk and dairy	-0.002***	-0.002***	-0.002***	-0.002***	-0.003***	-0.004***	-0.001*
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
	[100%]	[100%]	[110%]	[110%]	[90%]	[140%]	[100%]
Eggs	-0.012***	-0.014***	-0.013***	-0.013***	-0.012***	-0.016***	-0.010***
	(0.004)	(0.004)	(0.003)	(0.003)	(0.003)	(0.004)	(0.003)
	[370%]	[400%]	[390%]	[390%]	[240%]	[290%]	[310%]
Fish	-0.008*	-0.009*	-0.008*	-0.007	-0.006	-0.007	-0.004
	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)
	[80%]	[90%]	[80%]	[80%]	[50%]	[60%]	[80%]
All vegetables	-0.007	-0.019	-0.007	-0.036	-0.001	0.088	-0.044
	(0.060)	(0.061)	(0.062)	(0.071)	(0.066)	(0.057)	(0.068)
	[70%]	[70%]	[<5%]	[80%]	[30%]	[<5%]	[70%]
Roots, tubers, & plantain	0.016	0.004	0.016	0.004	0.034	0.111***	-0.021
	(0.051)	(0.050)	(0.052)	(0.059)	(0.055)	(0.047)	(0.057)
	[<5%]	[<5%]	[<5%]	[<5%]	[<5%]	[<5%]	[<5%]
Other leafy vegetables	-0.012	-0.010	-0.010	-0.017	-0.017	-0.012	-0.011
	(0.014)	(0.014)	(0.013)	(0.015)	(0.015)	(0.014)	(0.012)
	[110%]	[130%]	[120%]	[120%]	[80%]	[80%]	[110%]
Fruits	-0.002	-0.002	-0.002	-0.000	-0.000	-0.011	0.005
	(0.009)	(0.009)	(0.009)	(0.009)	(0.010)	(0.009)	(0.009)
	[80%]	[80%]	[80%]	[70%]	[50%]	[80%]	[80%]
Cereals, flour & bread	0.012	0.012	0.010	0.007	-0.002	-0.010	-0.010
	(0.020)	(0.019)	(0.020)	(0.020)	(0.024)	(0.021)	(0.014)
	[<5%]	[60%]	[<5%]	[<5%]	[30%]	[50%]	[<5%]
Pulse, nuts, & seeds	0.004	-0.002	0.004	0.004	0.002	0.000	0.005
	(0.006)	(0.005)	(0.006)	(0.005)	(0.006)	(0.006)	(0.005)
	[<5%]	[70%]	[<5%]	[<5%]	[<5%]	[<5%]	[<5%]
Oil and fat	0.003	0.004	0.003	0.004	0.003	0.002	0.006*
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.004)
	[<5%]	[<5%]	[<5%]	[<5%]	[<5%]	[<5%]	[<5%]
Spices	0.001	0.000	0.001	0.000	-0.001	-0.000	0.000
	(0.001)	(0.002)	(0.001)	(0.002)	(0.002)	(0.002)	(0.002)
	[<5%]	[70%]	[<5%]	[<5%]	[60%]	[60%]	[<5%]
Sugars and sweets	0.002	0.002	0.002	0.001	0.001	0.000	0.002
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
	[<5%]	[<5%]	[<5%]	[<5%]	[<5%]	[<5%]	[<5%]
Alcohol beverages	0.001	0.001	0.001	0.001	0.001	-0.002	-0.001
	(0.002)	(0.003)	(0.002)	(0.002)	(0.003)	(0.002)	(0.001)
	[<5%]	[<5%]	[<5%]	[<5%]	[<5%]	[190%]	[<5%]
Nonalcoholic beverages	-0.003	-0.004	-0.003	-0.004	-0.005	-0.009***	-0.000
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
	[130%]	[150%]	[140%]	[140%]	[100%]	[170%]	[130%]
N	8072	7477	8072	8129	8129	8129	8072

Notes: N = number of observations. Standard errors are in parentheses, and the maximum Γ is reported in square brackets. Columns follow the same robustness structure as earlier tables. *** p<0.01, ** p<0.05, * p<0.1.

Table D3: Placebo tests – Causal impacts of other broken appliances on food quantity

Food quantity category	(1) Broken fan (N=7,105)	(2) Broken radio (N=3,625)	(3) Broken sewing machine (N=1,605)
Total at home (excl. alcohol)	-0.098	-0.248*	0.147
	(0.149)	(0.142)	(0.182)
All meats	-0.008*	0.006	-0.004
	(0.004)	(0.006)	(0.005)
Milk & dairy products	-0.002	0.000	-0.001
	(0.001)	(0.002)	(0.002)
Eggs	-0.009	0.001	-0.003
	(0.006)	(0.008)	(0.008)
Fish	-0.000	0.001	0.013
	(0.009)	(0.007)	(0.011)
All vegetables	-0.029	-0.096	0.133
	(0.101)	(0.099)	(0.140)
Roots, tubers, & plantain	-0.019	-0.054	0.117
	(0.077)	(0.082)	(0.123)
Other leafy vegetables	0.005	-0.009	0.033
	(0.031)	(0.022)	(0.030)
Fruits	0.006	-0.010	0.002
	(0.019)	(0.013)	(0.016)
Cereals, flour & bread	0.012	-0.060***	-0.008
	(0.031)	(0.027)	(0.031)
Pulse, nuts, & seeds	-0.007	-0.006	0.007
	(0.008)	(0.009)	(0.013)
Oil & fats	0.011*	-0.005	0.007
	(0.006)	(0.004)	(0.006)
Spices	0.000	-0.001	0.000
	(0.003)	(0.002)	(0.003)
Sugar & sweets	-0.001	-0.001	-0.002
	(0.003)	(0.003)	(0.003)
Alcoholic beverages	0.004	-0.001	0.001
	(0.005)	(0.003)	(0.005)
Non-alcoholic beverages	-0.004	-0.003	-0.001
	(0.005)	(0.003)	(0.005)

Notes: N denotes the number of observations in each sample category. Columns (1), (2), and (3) report the ATT of having a broken fan, radio, and sewing machine, respectively, on food quantities. Standard errors are in parentheses. As discussed in the main text and shown in Appendices A2 and A3, TELASSO converged for broken fans only in the pooled sample due to the larger sample size. Therefore, fan is included as a placebo only in the pooled analysis and not in the food insecurity analysis, based solely on GLSS VII. *** p<0.01, ** p<0.05, * p<0.1.

Table D4: Gift sample – Impact of refrigerator breakdowns on food quantities among households that received the appliance as a gift

	(1)	(2)	(3)	(4)	(5)	(6)
Total at home (excl. alcohol)	0.105 (0.307)	0.084 (0.303)	0.086 (0.286)	0.027 (0.285)	0.044 (0.295)	0.004 (0.516)
All meats	-0.026*** (0.005)	-0.025*** (0.004)	-0.021*** (0.005)	-0.019*** (0.004)	-0.019*** (0.004)	-0.018*** (0.006)
Milk & dairy products	-0.007*** (0.002)	-0.006*** (0.002)	-0.006*** (0.002)	-0.004** (0.002)	-0.003 (0.002)	-0.002 (0.003)
Eggs	-0.017* (0.010)	-0.017* (0.010)	-0.016 (0.011)	-0.015 (0.011)	-0.012 (0.011)	-0.028 (0.018)
Fish	-0.024*** (0.009)	-0.023** (0.009)	-0.021** (0.010)	-0.017* (0.010)	-0.015 (0.010)	-0.022* (0.013)
All vegetables	0.138 (0.184)	0.104 (0.192)	0.129 (0.169)	0.058 (0.165)	0.051 (0.174)	0.042 (0.284)
Roots, tubers & plantain	0.160 (0.178)	0.134 (0.185)	0.166 (0.162)	0.102 (0.156)	0.098 (0.166)	0.075 (0.276)
Other leafy vegetables	-0.011 (0.026)	-0.015 (0.027)	-0.015 (0.029)	-0.018 (0.033)	-0.021 (0.031)	-0.020 (0.050)
Fruits	-0.016 (0.019)	-0.014 (0.019)	-0.009 (0.018)	-0.008 (0.019)	-0.007 (0.018)	-0.013 (0.024)
Cereals, flour & bread	-0.008 (0.071)	-0.005 (0.069)	-0.032 (0.064)	-0.036 (0.064)	-0.030 (0.065)	-0.093 (0.120)
Pulse, nuts, & seeds	0.017 (0.022)	0.017 (0.022)	0.014 (0.020)	0.015 (0.020)	0.014 (0.021)	0.023 (0.035)
Oil & fats	-0.003 (0.004)	-0.003 (0.004)	-0.003 (0.004)	-0.001 (0.004)	-0.000 (0.004)	-0.003 (0.007)
Spices	-0.001 (0.004)	-0.001 (0.004)	-0.002 (0.004)	-0.002 (0.004)	-0.002 (0.003)	-0.001 (0.004)
Sugar & sweets	0.000 (0.002)	0.001 (0.002)	0.000 (0.002)	0.000 (0.003)	-0.000 (0.003)	-0.002 (0.002)
Alcoholic beverages	-0.003* (0.002)	-0.002 (0.002)	-0.002 (0.002)	-0.002 (0.002)	-0.002 (0.002)	-0.001 (0.004)
Non-alcoholic beverages	-0.007 (0.006)	-0.007 (0.006)	-0.005 (0.006)	-0.004 (0.006)	-0.004 (0.006)	-0.004 (0.007)
Controls						
Age of refrigerator		✓	✓	✓	✓	✓
Region			✓	✓	✓	✓
Rural/urban				✓	✓	✓
Household variables					✓	✓
Community variables						✓
Observations	626	626	626	626	626	299

Notes: All specifications include interview year and interview month fixed effects. Observations include only households that received the refrigerator as a gift, using pooled GLSS VI and GLSS VII samples. Household variables include age and sex of the household head, schooling, employment status, household size, and dwelling characteristics (floor, roof, and wall materials). Community variables include access to markets, financial institutions, roads, living conditions, and main economic activity. Robust standard errors are reported in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

D.3 Food expenditure: summary statistics, model diagnostics and complementary results

Table D5: Summary statistics of food expenditure outcomes

Food expenditure category	(1) Functioning fridge (N=7,639)	(2) Broken fridge (N=490)	(3) Diff.	(4) SE
Total at home (excl. alcohol)	7.61	6.95	0.66**	0.27
All meats	0.76	0.52	0.25***	0.04
Milk & dairy products	0.22	0.15	0.07***	0.01
Eggs	0.13	0.09	0.04***	0.01
Fish	1.35	1.23	0.12**	0.06
All vegetables	2.00	2.13	-0.13	0.09
Roots, tubers, & plantain	1.00	1.21	-0.21***	0.06
Other leafy vegetables	0.98	0.90	0.08**	0.04
Fruits	0.30	0.23	0.07***	0.02
Cereals, flour & bread	1.53	1.44	0.09	0.06
Pulse, nuts, & seeds	0.16	0.18	-0.03**	0.01
Oil & fats	0.26	0.26	0.00	0.01
Spices	0.10	0.11	-0.01*	0.01
Sugar & sweets	0.16	0.13	0.03**	0.01
Alcoholic beverages	0.11	0.11	0.00	0.02
Non-alcoholic beverages	0.43	0.30	0.13***	0.03
Total out of home	1.56	1.25	0.32***	0.11
Hotels, cafes, restaurants	0.39	0.25	0.14***	0.05
Canteens	1.12	0.97	0.14	0.10

Notes: Data from GLSS VI and GLSS VII, pooled. Outcomes are reported as real daily food expenditure per adult equivalent (2017 GHS). In 2017, GHS 1 = USD 0.23 on average (Bank of Ghana, 2024). SE denotes the standard error of the mean difference. *** p<0.01, ** p<0.05, * p<0.10.

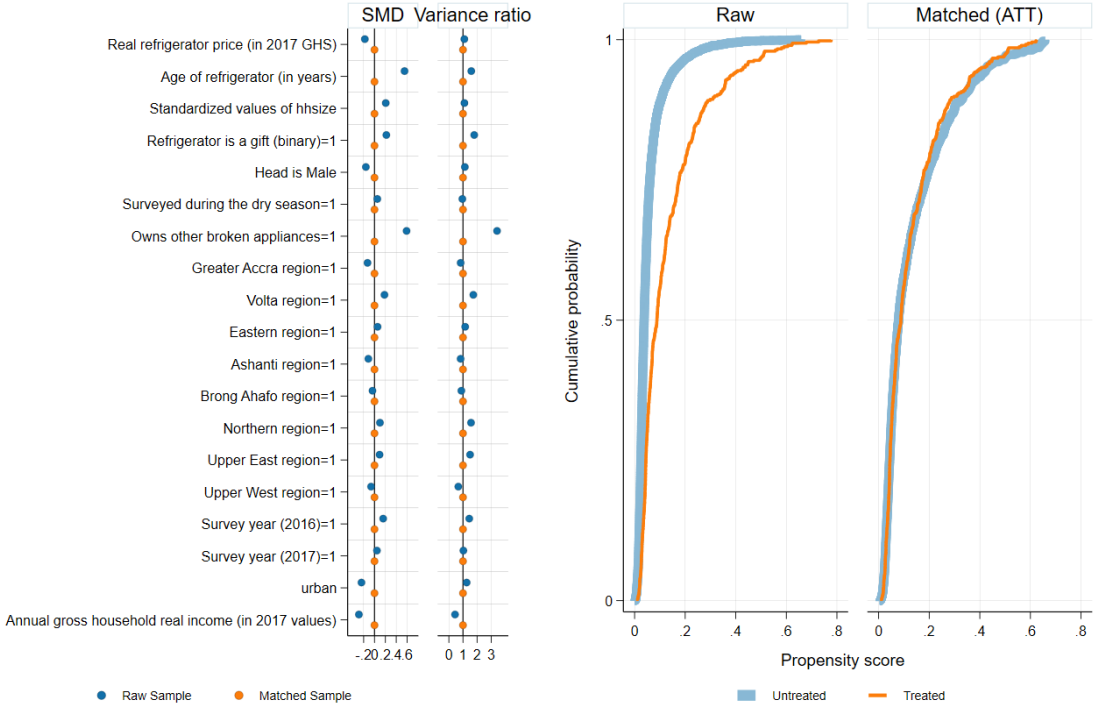


Figure D2: Model diagnostics for the food expenditure outcomes

Notes: Diagnostic tests when running our model on food expenditure outcomes, using pooled sample. The left panel presents the covariate balance using the standardized mean difference (SMD) and variance ratio. In the raw sample, we observe large disparities, even beyond 20% in the SMD and outside the recommended range in the variance ratio. However, in the matched sample and under the common support, the SMD is zero (0) and the variance ratio is one (1) for all covariates. The right panel presents the cumulative probability plots of the propensity scores before matching (Raw) and after matching (Matched) under the common support. ATT = Average Treated effect on the Treated. The vertical axis of the right panel reports the probability that the score is less than or equal to a given propensity score.

Table D6: Causal impacts of having a broken refrigerator on food expenditure

Food expenditure category	(1) ATT	(2) Treated group mean	(3) Max. Γ	(4) Entropy balance
Total at home (excl. alcohol)	-0.065	6.95	50%	-0.107
	(0.285)			(0.284)
All meats	-0.141***	0.52	180%	-0.139***
	(0.044)			(0.039)
Milk & dairy products	-0.023***	0.15	100%	-0.032***
	(0.011)			(0.011)
Eggs	-0.019***	0.09	120%	-0.021***
	(0.007)			(0.007)
Fish	-0.053	1.23	70%	-0.064
	(0.055)			(0.059)
All vegetables	0.101	2.13	<5%	0.115
	(0.109)			(0.109)
Roots, tubers, & plantain	0.144*	1.21	<5%	0.153*
	(0.082)			(0.082)
Other leafy vegetables	-0.039	0.90	70%	-0.034
	(0.044)			(0.041)
Fruits	-0.004	0.23	<5%	-0.000
	(0.018)			(0.018)
Cereals, flour & bread	0.031	1.44	<5%	0.005
	(0.063)			(0.065)
Pulse, nuts, & seeds	0.024	0.18	<5%	0.024
	(0.016)			(0.015)
Oil & fats	0.012	0.26	<5%	0.011
	(0.015)			(0.014)
Spices	0.010*	0.11	<5%	0.008
	(0.006)			(0.006)
Sugar & sweets	-0.007	0.13	90%	-0.004
	(0.010)			(0.009)
Alcoholic beverages	0.027*	0.11	<5%	0.024
	(0.016)			(0.016)
Non-alcoholic beverages	-0.025	0.30	90%	-0.027
	(0.022)			(0.021)
Total out of home	0.172*	1.25	<5%	0.134
	(0.091)			(0.103)
Hotels, cafes, restaurants	0.010	0.25	130%	0.003
	(0.031)			(0.032)
Canteens	0.159*	0.97	<5%	0.130
	(0.088)			(0.098)
N	8072			8129

Notes: N is the total number of households in the sample category. Standard errors are in parentheses. Column (1) reports the ATT from TELASSO. Column (2) reports the mean of the treated group. Column (3) reports the maximum Γ (in percent) from the Rosenbaum bounds sensitivity analysis at the 5% critical level. Column (4) reports the corresponding ATT from entropy balance estimation using the same preselected and LASSO-selected covariates as in the baseline. *** p<0.01, ** p<0.05, * p<0.10.

Table D7: Alternative specifications for food expenditure

Food expenditure category	Extra covariates	Excluding gifts	Adaptive LASSO	IPW	PSM	MD	Survey weights
Total at home (excl. alcohol)	-0.047	-0.220	-0.062	-0.136	-0.262	-0.351	-0.031
	(0.334)	(0.327)	(0.331)	(0.333)	(0.341)	(0.345)	(0.294)
	[<5%]	[50%]	[50%]	[50%]	[40%]	[60%]	[<5%]
All meat	-0.176***	-0.146***	-0.172***	-0.192***	-0.236***	-0.222***	-0.146***
	(0.058)	(0.059)	(0.058)	(0.053)	(0.066)	(0.050)	(0.048)
	[150%]	[160%]	[180%]	[190%]	[130%]	[190%]	[110%]
Milk and dairy	-0.027***	-0.025*	-0.027***	-0.034***	-0.052***	-0.065***	-0.018
	(0.013)	(0.013)	(0.013)	(0.013)	(0.013)	(0.013)	(0.012)
	[90%]	[100%]	[100%]	[100%]	[100%]	[150%]	[<5%]
Eggs	-0.022***	-0.020***	-0.021***	-0.027***	-0.022***	-0.041***	-0.018***
	(0.009)	(0.009)	(0.009)	(0.009)	(0.009)	(0.009)	(0.007)
	[120%]	[120%]	[120%]	[130%]	[100%]	[160%]	[60%]
Fish	-0.086	-0.137***	-0.085	-0.100	-0.108	-0.085	-0.046
	(0.066)	(0.065)	(0.065)	(0.069)	(0.067)	(0.069)	(0.055)
	[60%]	[60%]	[70%]	[70%]	[40%]	[60%]	[<5%]
All vegetables	0.199	0.169	0.183	0.187	0.181	0.328***	0.124
	(0.156)	(0.163)	(0.156)	(0.155)	(0.158)	(0.155)	(0.166)
	[<5%]	[<5%]	[<5%]	[<5%]	[<5%]	[<5%]	[<5%]
Roots, tubers, & plantain	0.247*	0.187	0.233*	0.232*	0.228*	0.361***	0.167
	(0.129)	(0.136)	(0.129)	(0.129)	(0.129)	(0.129)	(0.153)
	[<5%]	[<5%]	[<5%]	[<5%]	[<5%]	[<5%]	[<5%]
Other leafy vegetables	-0.044	-0.023	-0.042	-0.045	-0.047	-0.033	-0.042
	(0.056)	(0.056)	(0.055)	(0.055)	(0.057)	(0.053)	(0.042)
	[60%]	[60%]	[70%]	[70%]	[40%]	[50%]	[<5%]
Fruits	-0.003	-0.010	-0.003	0.001	0.002	-0.031	0.035
	(0.029)	(0.034)	(0.029)	(0.028)	(0.030)	(0.029)	(0.043)
	[<5%]	[90%]	[<5%]	[<5%]	[<5%]	[100%]	[<5%]
Cereals, flour & bread	0.041	-0.005	0.040	0.015	-0.013	-0.167	0.000
	(0.086)	(0.067)	(0.085)	(0.087)	(0.092)	(0.116)	(0.058)
	[<5%]	[40%]	[<5%]	[<5%]	[20%]	[60%]	[<5%]
Pulse, nuts, & seeds	0.070	0.005	0.070	0.074	0.068	0.067	0.041*
	(0.056)	(0.017)	(0.056)	(0.056)	(0.059)	(0.056)	(0.025)
	[<5%]	[<5%]	[<5%]	[<5%]	[<5%]	[<5%]	[<5%]
Oil and fat	0.006	0.008	0.004	0.005	0.004	-0.004	0.022
	(0.016)	(0.016)	(0.016)	(0.015)	(0.016)	(0.015)	(0.016)
	[<5%]	[<5%]	[<5%]	[<5%]	[<5%]	[30%]	[<5%]
Spices	0.011	0.014	0.010	0.006	0.002	0.006	0.004
	(0.008)	(0.008)	(0.008)	(0.008)	(0.009)	(0.008)	(0.007)
	[<5%]	[<5%]	[<5%]	[<5%]	[<5%]	[<5%]	[<5%]
Sugars and sweets	-0.014	-0.011	-0.014	-0.014	-0.022*	-0.028***	-0.002
	(0.012)	(0.013)	(0.012)	(0.011)	(0.013)	(0.011)	(0.010)
	[80%]	[80%]	[90%]	[90%]	[60%]	[90%]	[<5%]
Alcohol beverages	0.037	0.047	0.036	0.030	0.049	0.001	0.000
	(0.040)	(0.047)	(0.040)	(0.039)	(0.039)	(0.044)	(0.031)
	[<5%]	[<5%]	[<5%]	[<5%]	[<5%]	[<5%]	[<5%]
Nonalcoholic beverages	-0.052***	-0.060***	-0.051***	-0.057***	-0.068***	-0.109***	-0.018
	(0.024)	(0.025)	(0.023)	(0.023)	(0.027)	(0.023)	(0.021)
	[90%]	[90%]	[100%]	[100%]	[60%]	[130%]	[<5%]
Total out of home	0.197*	0.183	0.190*	0.162	0.193	-0.149	0.082
	(0.106)	(0.111)	(0.106)	(0.117)	(0.125)	(0.123)	(0.083)
	[<5%]	[<5%]	[<5%]	[<5%]	[<5%]	[110%]	[<5%]
Hotels, cafes, & restaurants	-0.025	-0.039	-0.027	-0.026	-0.012	-0.111***	-0.026
	(0.040)	(0.045)	(0.040)	(0.039)	(0.042)	(0.041)	(0.028)
	[130%]	[150%]	[130%]	[130%]	[80%]	[170%]	[<5%]
Canteens	0.213***	0.207*	0.218***	0.189	0.205*	-0.038	0.109
	(0.108)	(0.111)	(0.105)	(0.115)	(0.123)	(0.121)	(0.079)
	[<5%]	[<5%]	[<5%]	[<5%]	[<5%]	[100%]	[<5%]
N	8072	7477	8072	8129	8129	8129	8072

Notes: N = number of observations. Standard errors are in parentheses, and the maximum Γ is reported in square brackets. Columns follow the same robustness structure as earlier appendix tables. *** p<0.01, ** p<0.05, * p<0.1.

Table D8: Placebo – Causal impacts of other broken appliances on food expenditure

Food expenditure category	(1) Broken fan (N=7,105)	(2) Broken radio (N=3,625)	(3) Broken sewing machine (N=1,605)
Total at home (excl. alcohol)	0.12	0.19	0.55
	(0.57)	(0.46)	(0.57)
All meats	-0.07	0.05	0.06
	(0.08)	(0.08)	(0.10)
Milk & dairy products	-0.02	0.02	0.00
	(0.02)	(0.02)	(0.03)
Eggs	-0.00	-0.01	-0.01
	(0.01)	(0.01)	(0.01)
Fish	-0.05	0.03	0.12
	(0.10)	(0.09)	(0.11)
All vegetables	0.17	0.07	0.38*
	(0.23)	(0.18)	(0.22)
Roots, tubers, & plantain	0.06	0.07	0.26
	(0.14)	(0.14)	(0.17)
Other leafy vegetables	0.06	0.03	0.12
	(0.08)	(0.06)	(0.08)
Fruits	0.01	-0.00	0.05
	(0.04)	(0.03)	(0.04)
Cereals, flour & bread	0.01	-0.07	0.05
	(0.11)	(0.10)	(0.12)
Pulse, nuts, & seeds	-0.02	0.03	0.01
	(0.02)	(0.03)	(0.04)
Oil & fats	0.03	-0.03	-0.01
	(0.03)	(0.02)	(0.02)
Spices	0.01	0.01	0.00
	(0.01)	(0.01)	(0.01)
Sugar & sweets	-0.01	0.00	-0.01
	(0.02)	(0.02)	(0.02)
Alcoholic beverages	0.06*	0.02	0.01
	(0.04)	(0.03)	(0.02)
Non-alcoholic beverages	-0.05	0.01	0.01
	(0.04)	(0.03)	(0.04)
Total out of home	-0.23***	0.15	-0.11
	(0.11)	(0.15)	(0.08)
Hotels, cafes, restaurants	0.06	-0.03	0.00
	(0.08)	(0.05)	(0.05)
Canteens	-0.28***	0.15	-0.11
	(0.08)	(0.14)	(0.07)

Notes: N denotes the total number of households in each sample category. Columns (1), (2), and (3) report the ATT of having a broken fan, radio, and sewing machine on food expenditure, respectively. Standard errors are in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

E Micronutrients: summary statistics and average treatment effects

Table E1: Mean statistics of all essential nutrients by refrigerator functionality status

	Animal sources				Plant sources			
Nutrient intake (unit/day/AE)	Functioning fridge	Broken fridge	Diff.	SE	Functioning fridge	Broken fridge	Diff.	SE
Metabolizable energy (kcal)	282	196	86.9***	15.0	2313	2448	-134	113
Carbohydrate (g)	2.78	1.78	1.00***	0.17	390	418	-28.5	20.3
Protein (g)	36.9	27.2	9.66***	1.97	39.0	42.7	-3.66*	2.13
Fat (g)	13.7	8.73	4.94***	0.81	62.7	64.8	-2.12	3.22
Saturated fat (g)	5.23	3.35	1.88***	0.29	25.3	24.8	0.53	1.71
Monounsaturated fat (g)	5.17	3.29	1.88***	0.32	14.1	15.4	-1.24	0.82
Polyunsaturated fat (g)	1.60	1.07	0.53***	0.11	11.3	12.6	-1.23*	0.67
Cholesterol (g)	0.30	0.19	0.10***	0.02	–	–	–	–
Minerals (g)	1.74	1.24	0.50***	0.10	8.91	9.0	-0.08	0.46
Iron (mg)	3.55	2.54	1.01***	0.22	24.3	27.2	-2.99***	1.15
Zinc (mg)	3.27	2.24	1.03***	0.19	12.2	14.5	-2.33***	0.78
Calcium (mg)	356	257	99.1***	28.2	323	343	-20.0	17.7
Phosphorous (g)	0.58	0.41	0.17***	0.04	0.96	1.03	-0.07	0.05
Potassium (g)	0.51	0.37	0.14***	0.03	3.55	3.7	-0.11	0.19
Magnesium (mg)	55.3	41.7	13.6***	2.77	398	433	-35.0	21.3
Copper (mg)	0.20	0.15	0.05***	0.01	2.64	3.40	-0.75***	0.28
Vitamins	18.1	13.3	4.73***	0.93	255	269	-14.6	14.6
Vitamin A (RAE) (mg)	0.09	0.06	0.03***	0.01	0.93	0.95	-0.02	0.07
Retinol (mcg)	84.5	54.0	30.5***	5.77	–	–	–	–
Beta-carotene equivalents (mg)	–	–	–	–	10.9	11.3	-0.33	0.81
Alpha-carotene (mg)	–	–	–	–	0.87	0.83	0.04	0.08
Beta-carotene (mg)	–	–	–	–	10.0	10.5	-0.45	0.78
Vitamin B1 (mcg)	166	106	59.5***	10.6	1634	2429	-795***	257
Vitamin B2 (mg)	0.27	0.19	0.08***	0.01	1.96	2.81	-0.85***	0.27
Vitamin B3 (mg)	13.6	9.95	3.60***	0.70	17.2	18.6	-1.38	0.88
Vitamin B6 (mg)	0.49	0.35	0.14***	0.03	2.30	3.04	-0.74**	0.27
Vitamin B9 (mg)	0.03	0.02	0.01***	0.00	0.38	0.42	-0.04**	0.02
Vitamin B12 (mcg)	8.13	5.84	2.28***	0.79	–	–	–	–
Vitamin C (mg)	0.51	0.42	0.09*	0.05	222	232	-10.00	13.3
Vitamin D (mcg)	5.04	3.40	1.64***	0.35	0.63	1.35	-0.72***	0.25
Vitamin E (mg)	2.93	2.21	0.72***	0.18	8.26	9.12	-0.87*	0.47
Fibre (g)	–	–	–	–	52.7	58.9	-6.19**	2.96

Notes: Pooled data from GLSS VI and GLSS VII matched with the West Africa Food Composition Table. N denotes total sample in each category. SE = standard error of the mean difference. Animal sources include all meats, eggs, fish, and milk & dairy products. Plant sources include all remaining food groups. Vitamin A (RAE) is retinol activity equivalent. See Vincent et al. (2020) for detailed computational definitions. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table E2: ATT of refrigeration on home-based micronutrient intake (excl. alcohol)

	Animal origin		Plant based		Total (excl. alcohol)	
Nutrient intake	ATT	Max. Γ	ATT	Max. Γ	ATT	Max. Γ
Minerals						
Iron	-0.56***	100%	1.997	<5%	1.433	<5%
	(0.175)		(1.403)		(1.472)	
Zinc	-0.589***	160%	1.125	<5%	0.526	<5%
	(0.153)		(0.961)		(1.014)	
Calcium	-57.36***	150%	13.605	<5%	-43.774	<5%
	(20.211)		(19.404)		(31.556)	
Phosphorous	-0.09***	170%	0.069	<5%	-0.027	<5%
	(0.027)		(0.058)		(0.072)	
Potassium	-0.08***	140%	0.084	<5%	0.000	<5%
	(0.022)		(0.200)		(0.211)	
Magnesium	-6.57***	110%	29.875	<5%	22.691	<5%
	(2.377)		(24.514)		(25.644)	
Copper	-0.02	110%	0.328	<5%	0.312	<5%
	(0.013)		(0.383)		(0.384)	
Vitamins						
Vitamin A (RAE)	-0.017***	150%	0.046	<5%	0.029	<5%
	(0.005)		(0.076)		(0.078)	
Retinol (mcg)	-16.5***	150%	–	–	-16.49***	150%
	(4.354)		–		(4.354)	
Beta-carotene equiv. (mg)	–	–	0.689	<5%	0.689	<5%
	–		(0.900)		(0.900)	
Alpha-carotene (mg)	–	–	-0.041	70%	-0.041	70%
	–		(0.081)		(0.081)	
Beta-carotene (mg)	–	–	0.733	<5%	0.733	<5%
	–		(0.878)		(0.878)	
Vitamin B1 (mg)	-31.91***	170%	318.373	<5%	286.42	<5%
	(8.396)		(364.840)		(365.58)	
Vitamin B2	-0.042***	110%	0.340	<5%	0.299	<5%
	(0.011)		(0.367)		(0.368)	
Vitamin B3	-1.975***	100%	1.112	<5%	-0.909	50%
	(0.594)		(0.998)		(1.351)	
Vitamin B6 (mg)	-0.077***	110%	0.281	<5%	0.204	<5%
	(0.022)		(0.376)		(0.380)	
Vitamin B9	-0.007***	310%	0.031	<5%	0.024	<5%
	(0.002)		(0.024)		(0.025)	
Vitamin B12 (mcg)	-1.474***	150%	–	–	-1.47***	150%
	(0.583)		–		(0.583)	
Vitamin C (mg)	0.017	<5%	6.507	<5%	6.526	<5%
	(0.050)		(15.013)		(15.017)	
Vitamin D (mcg)	-0.982***	150%	0.254	<5%	-0.715	130%
	(0.272)		(0.355)		(0.444)	
Vitamin E (mg)	-0.383***	120%	0.703	<5%	0.318	<5%
	(0.160)		(0.551)		(0.627)	

Notes: $N = 8072$. Standard errors are in parentheses. Beta-carotene and alpha-carotene are provitamin A. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

F Comparing households with vs. without refrigerators

Before this study, some analyses have compared households that own a refrigerator with those that do not. Here, we replicate the baseline specification but redefine the treatment to compare households with a functioning refrigerator (treatment) to households without a refrigerator or with a broken one (control).

The resulting ATTs are reported in Tables F1, F2, and F3. While the estimates for food insecurity are broadly similar to the baseline results, they rely on a substantially stronger identifying assumption than the one maintained in the core analysis. Moreover, the estimated effects on food quantities and expenditures differ markedly from the baseline and are not robust to hidden bias.

Table F1: Impacts of having a refrigerator on food insecurity

Food insecurity outcomes	(1) ATT	(2) Treated group mean	(3) Max. Γ
Risk of food insecurity	-0.17***	0.38	>400%
	(0.02)		
Food insecurity intensity	-1.05***	1.55	160%
	(0.08)		
Worried about not having enough food to eat	-0.16***	0.29	<5%
	(0.01)		
Unable to eat healthy and nutritious foods	-0.17***	0.24	30%
	(0.01)		
Ate only a few kinds of foods	-0.18***	0.27	<10%
	(0.01)		
Had to skip a meal	-0.15***	0.21	50%
	(0.01)		
Ate less than the respondent's expectation	-0.16***	0.22	40%
	(0.01)		
Ran out of food	-0.13***	0.18	100%
	(0.01)		
Went without eating for a whole day	-0.10***	0.11	260%
	(0.01)		
Hungry but did not eat	-0.03***	0.03	>400%
	(0.01)		

Notes: Total number of households = 13,774; and the treated households are 3,455. The treated group consists of households with a functioning refrigerator. Column (1) reports the average treatment effect on the treated (ATT). Column (2) reports the mean of the treated group. Column (3) reports the maximum Γ (in percent) from the Rosenbaum bounds sensitivity analysis at the 5% critical level. Standard errors are in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table F2: Impacts of having a refrigerator on food quantity

Food expenditure category	(1) ATT	(2) Treated group mean	(3) Max. Γ
Total at home (excl. alcohol)	0.206***	1.69	<5%
	(0.037)		
All Meats	0.017***	0.04	10%
	(0.001)		
Milk & dairy products	0.006***	0.01	10%
	(0.000)		
Eggs	0.016***	0.04	<5%
	(0.002)		
Fish	0.020***	0.08	<5%
	(0.002)		
All vegetables	0.029	0.80	<5%
	(0.023)		
Roots, tubers, & plantain	-0.013	0.54	<5%
	(0.018)		
Other leafy vegetables	0.036***	0.23	<5%
	(0.007)		
Fruits	0.036***	0.13	<5%
	(0.004)		
Cereals, flour & bread	0.028***	0.29	<5%
	(0.007)		
Pulse, nuts, & seeds	0.003	0.06	<5%
	(0.003)		
Oil & fats	0.008***	0.04	<5%
	(0.001)		
Spices	0.003***	0.02	<5%
	(0.001)		
Sugar & sweets	0.004***	0.02	<5%
	(0.001)		
Alcoholic beverages	0.002*	0.02	<5%
	(0.001)		
Non-alcoholic beverages	0.016***	0.04	<5%
	(0.001)		

Notes: Total number of households = 30,589; and the treated households are 7,859. The treated group consists of households with a functioning refrigerator. Column (1) reports the average treatment effect on the treated (ATT). Column (2) reports the mean of the treated group. Column (3) reports the maximum Γ (in percent) from the Rosenbaum bounds sensitivity analysis at the 5% critical level. Standard errors are in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table F3: Impacts of having a refrigerator on food expenditure

Food expenditure category	(1) ATT	(2) Treated group mean	(3) Max. Γ
Total at home (excl. alcohol)	1.65***	7.63	10%
	(0.12)		
All Meats	0.30***	0.76	20%
	(0.02)		
Milk & dairy products	0.09***	0.22	20%
	(0.01)		
Eggs	0.04***	0.13	<5%
	(0.00)		
Fish	0.27***	1.35	<5%
	(0.02)		
All vegetables	0.19***	2.02	<5%
	(0.04)		
Roots, tubers, & plantain	0.01	1.01	<5%
	(0.03)		
Other leafy vegetables	0.18***	0.98	<5%
	(0.02)		
Fruits	0.09***	0.30	<5%
	(0.01)		
Cereals, flour & bread	0.28***	1.53	<5%
	(0.03)		
Pulse, nuts, & seeds	0.01***	0.16	<5%
	(0.01)		
Oil & fats	0.06***	0.26	<5%
	(0.01)		
Spices	0.02***	0.10	<5%
	(0.00)		
Sugar & sweets	0.05***	0.16	<5%
	(0.00)		
Alcoholic beverages	0.01	0.11	<5%
	(0.01)		
Non-alcoholic beverages	0.14***	0.43	<5%
	(0.01)		
Total out of home	-0.08	1.55	100%
	(0.05)		
Hotels, cafes, & restaurants	0.04***	0.39	<5%
	(0.02)		
Canteens	-0.14***	1.10	150%
	(0.05)		

Notes: Total number of households = 30,589; and the treated households are 7,859. The treated group consists of households with a functioning refrigerator. Column (1) reports the average treatment effect on the treated (ATT). Column (2) reports the mean of the treated group. Column (3) reports the maximum Γ (in percent) from the Rosenbaum bounds sensitivity analysis at the 5% critical level. Standard errors are in parentheses. *** p<0.01, ** p<0.05, * p<0.10.